

# Notebook

April 4, 2026

## 1 Dask on Strato — Setup Guide

This guide picks up where the [CLAAUDIA getting-started guide](#) leaves off. By the end of that guide you have a running Ubuntu VM and can SSH into it. This guide takes you from there to a working multi-VM Dask cluster.

\*\*\*\*VPN required if off-campus.\*\* Strato is only reachable from the AAU network. On campus (group room WiFi or wired) you can reach Strato directly — no VPN needed. Working from home or elsewhere: connect via the AAU VPN first.\*\*

### 1.1 Part 1 — Configure the Head Node

#### 1.1.1 1.1 Open the extra ports in your security group

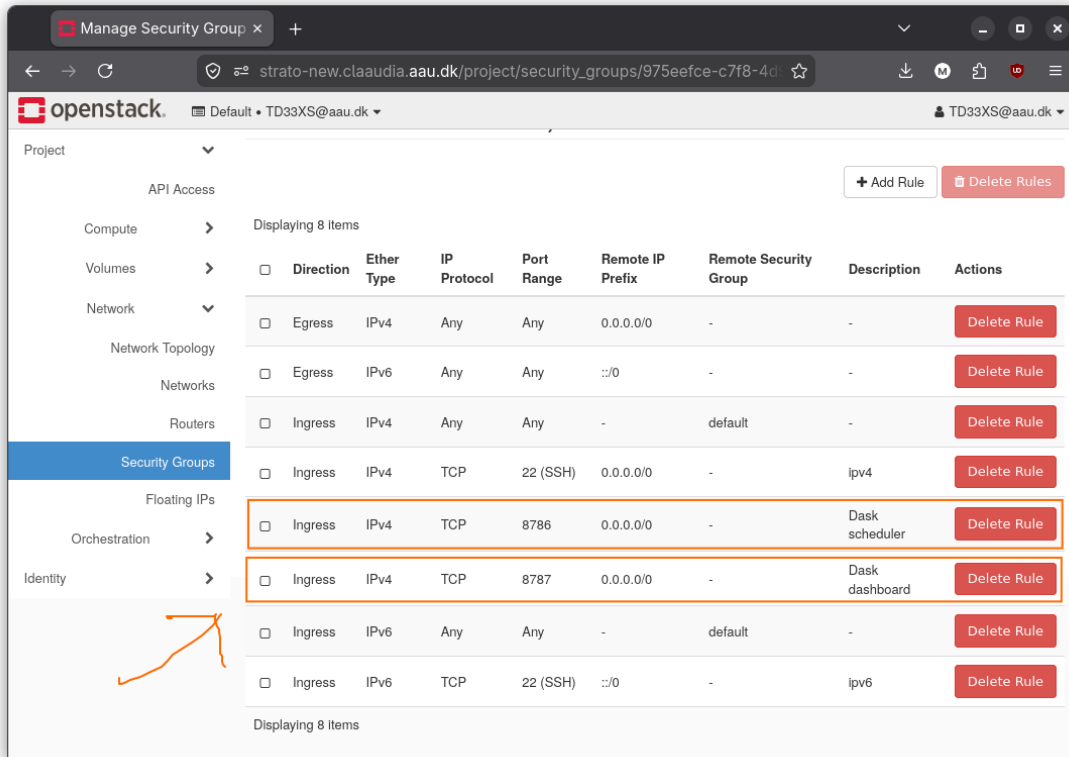
Dask needs two ports in addition to SSH:

1. In the Strato dashboard, go to Network → Security Groups.\*\*\*\*
2. Click Manage Rules\*\* next to your security group.\*\*
3. Add two rules — click Add Rule\*\* for each.\*\*

| Direction | Protocol | Port | Remote    |
|-----------|----------|------|-----------|
| Ingress   | TCP      | 8787 | 0.0.0.0/0 |
| Ingress   | TCP      | 8786 | 0.0.0.0/0 |

Port 8786 is the Dask scheduler; port 8787 is the dashboard.

\*\*\*\*Recommended flavor:\*\* AAU.CPU.b.2-4 (2 vCPUs, 4 GB RAM) for both head node and worker VMs. The CLAAUDIA getting-started guide uses AAU.CPU.a.1-4 (1 vCPU) — resize before continuing.\*\*

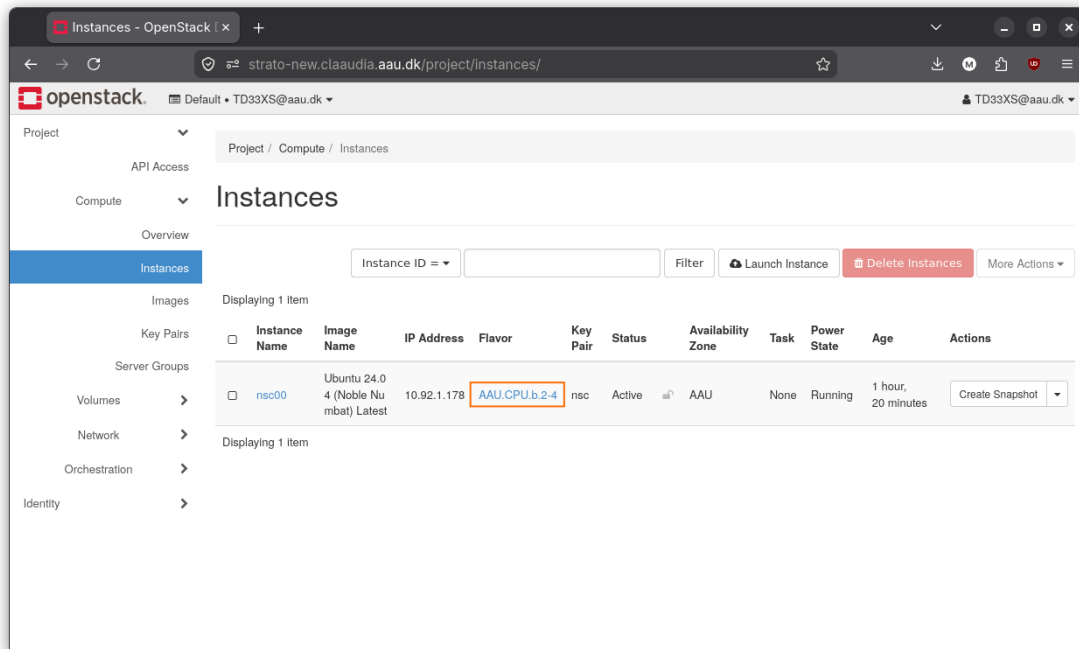


### 1.1.2 1.2 Resize the VM to AAU.CPU.b.2-4

The CLAUDIA getting-started guide launches a AAU.CPU.a.1-4 VM (1 vCPU). Resize it before proceeding — the head node runs both the Dask scheduler and your Python script, and 1 vCPU causes contention.

1. \*\*\*\*Shut off the VM first:\*\* in the Strato dashboard go to **Compute** → **Instances**, click the dropdown → **Shut Off Instance**. Wait until status is Shutoff.\*\*
2. **Click the dropdown again** → **Resize Instance**.
3. **Select AAU.CPU.b.2-4 (2 vCPUs, 4 GB RAM) and click Resize**.
4. **Once the status changes to Verify Resize, click the dropdown** → **Confirm Resize**.
5. **Start the VM again: dropdown** → **Start Instance**. **Wait until Active**.

I created the VM using AAU.CPU.b.2-4 instead of resizing it.



### 1.1.3 1.3 SSH into your VM

Use the IP shown in Compute → Instances\*\* next to your VM (e.g. 10.92.y.x):\*\*

```
ssh -i ~/.ssh/my_ssh_key ubuntu@10.92.y.x
```

All commands from here until Part 3 are run on the VM, not on your laptop.

I get the following output:

```
(mikkel@pella)-[~/Sync/AAU/Semester 8/github/nsc-mikkel]
$ ssh -i nsc ubuntu@10.92.1.178
The authenticity of host '10.92.1.178 (10.92.1.178)' can't be established.
ED25519 key fingerprint is: SHA256:/rPURLMrP70tghOKRq/g/CPC0Td6PZr8uD3UXue010M
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.92.1.178' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)
```

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

System information as of Thu Mar 26 12:57:49 UTC 2026

```
System load: 0.11          Processes:              118
Usage of /:   3.4% of 47.39GB Users logged in:        0
Memory usage: 5%          IPv4 address for ens3: 10.92.1.178
Swap usage:   0%
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.  
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.  
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".  
See "`man sudo_root`" for details.

ubuntu@nsc00:~\$

#### 1.1.4 1.4 Install Miniconda

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh -b -p ~/miniconda3
~/miniconda3/bin/conda init bash
source ~/.bashrc
```

I get the following output:

```
ubuntu@nsc00:~$ wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
--2026-03-26 12:59:56-- https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 104.16.32.241, 104.16.191.158, 2606:4700::68
Connecting to repo.anaconda.com (repo.anaconda.com)|104.16.32.241|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 162067098 (155M) [application/octet-stream]
Saving to: 'Miniconda3-latest-Linux-x86_64.sh'
```

```
Miniconda3-latest-Linux-x86_64.sh 100%[=====
```

```
2026-03-26 12:59:57 (140 MB/s) - 'Miniconda3-latest-Linux-x86_64.sh' saved [162067098/162067098]
```

```
ubuntu@nsc00:~$ bash Miniconda3-latest-Linux-x86_64.sh -b -p ~/miniconda3
PREFIX=/home/ubuntu/miniconda3
Unpacking bootstrapper...
```

Unpacking payload...

Installing base environment...

Preparing transaction: ...working... done

Executing transaction: ...working... done

installation finished.

```
ubuntu@nsc00:~$ ~/miniconda3/bin/conda init bash
```

```
no change      /home/ubuntu/miniconda3/condabin/conda
```

```
no change      /home/ubuntu/miniconda3/bin/conda
```

```
no change      /home/ubuntu/miniconda3/bin/conda-env
```

```
no change      /home/ubuntu/miniconda3/bin/activate
```

```
no change      /home/ubuntu/miniconda3/bin/deactivate
```

```
no change      /home/ubuntu/miniconda3/etc/profile.d/conda.sh
```

```
no change      /home/ubuntu/miniconda3/etc/fish/conf.d/conda.fish
```

```
no change      /home/ubuntu/miniconda3/shell/condabin/Conda.psm1
```

```
no change      /home/ubuntu/miniconda3/shell/condabin/conda-hook.ps1
```

```
no change      /home/ubuntu/miniconda3/lib/python3.13/site-packages/xontrib/conda.xsh
```

```
no change      /home/ubuntu/miniconda3/etc/profile.d/conda.csh
```

```
modified      /home/ubuntu/.bashrc
```

==> For changes to take effect, close and re-open your current shell. <==

```
ubuntu@nsc00:~$ source ~/.bashrc
```

```
(base) ubuntu@nsc00:~$
```

Verify it worked:

```
conda --version
```

I get the following output:

```
(base) ubuntu@nsc00:~$ conda --version
```

```
conda 26.1.1
```

### 1.1.5 1.5 Create the Python environment

```
conda create -n nsc python=3.11 "dask[distributed]" numba numpy matplotlib -c conda-forge -y
conda activate nsc
```

If you see a `CondaToSNonInteractiveError` about Terms of Service, run these two lines exactly as shown, then re-run the conda create command above:

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main
```

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r
```

This takes a couple of minutes. When it finishes, verify the key packages. Make sure your prompt shows (nsc) before running — if it still shows (base), run `conda activate nsc` first:

```
python -c "import dask, numba, numpy; print(dask.__version__, numba.__version__, numpy.__version__)"
```

I get the following output:

```
(base) ubuntu@nsc00:~$ conda create -n nsc python=3.11 "dask[distributed]" numba numpy matplotlib
```

CondaToSNonInteractiveError: Terms of Service have not been accepted for the following channel:

- <https://repo.anaconda.com/pkg/main>
- <https://repo.anaconda.com/pkg/r>

To accept these channels' Terms of Service, run the following commands:

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/main
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/r
```

For information on safely removing channels from your conda configuration, please see the official documentation:

<https://www.anaconda.com/docs/tools/working-with-conda/channels>

```
(base) ubuntu@nsc00:~$ conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/main
accepted Terms of Service for https://repo.anaconda.com/pkg/main
```

```
(base) ubuntu@nsc00:~$ conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/r
accepted Terms of Service for https://repo.anaconda.com/pkg/r
```

Then I can install and activate:

```
(base) ubuntu@nsc00:~$ conda create -n nsc python=3.11 "dask[distributed]" numba numpy matplotlib
```

2 channel Terms of Service accepted

Retrieving notices: done

Channels:

- conda-forge
- defaults

Platform: linux-64

Collecting package metadata (repodata.json): done

Solving environment: done

*## Package Plan ##*

environment location: /home/ubuntu/miniconda3/envs/nsc

added / updated specs:

- dask
- matplotlib
- numba
- numpy
- python=3.11

The following packages will be downloaded:

| package | build |
|---------|-------|
| -----   | ----- |

|                                          |  |                 |        |             |
|------------------------------------------|--|-----------------|--------|-------------|
| _openmp_mutex-4.5                        |  | 20_gnu          | 28 KB  | conda-forge |
| alsa-lib-1.2.15.3                        |  | hb03c661_0      | 571 KB | conda-forge |
| aws-c-auth-0.10.1                        |  | h2d2dd48_2      | 131 KB | conda-forge |
| aws-c-cal-0.9.13                         |  | h2c9d079_1      | 55 KB  | conda-forge |
| aws-c-common-0.12.6                      |  | hb03c661_0      | 234 KB | conda-forge |
| aws-c-compression-0.3.2                  |  | h8b1a151_0      | 22 KB  | conda-forge |
| aws-c-event-stream-0.6.0                 |  | h9b893ba_1      | 58 KB  | conda-forge |
| aws-c-http-0.10.12                       |  | h4bacb7b_1      | 221 KB | conda-forge |
| aws-c-io-0.26.3                          |  | hc87160b_0      | 177 KB | conda-forge |
| aws-c-mqtt-0.15.2                        |  | he9ea9c5_1      | 217 KB | conda-forge |
| aws-c-s3-0.11.5                          |  | h6d69fc9_5      | 148 KB | conda-forge |
| aws-c-sdkutils-0.2.4                     |  | h8b1a151_4      | 58 KB  | conda-forge |
| aws-checksums-0.2.10                     |  | h8b1a151_0      | 99 KB  | conda-forge |
| aws-crt-cpp-0.37.4                       |  | h4c8aef7_3      | 401 KB | conda-forge |
| aws-sdk-cpp-1.11.747                     |  | hc3785e1_3      | 3.5 MB | conda-forge |
| azure-core-cpp-1.16.2                    |  | h206d751_0      | 341 KB | conda-forge |
| azure-identity-cpp-1.13.3                |  | hed0cdb0_1      | 245 KB | conda-forge |
| azure-storage-blobs-cpp-12.16.0          |  | hdd73cc9_1      | 566 KB | conda-forge |
| azure-storage-common-cpp-12.12.0         |  | ha7a2c86_1      | 147 KB | conda-forge |
| azure-storage-files-datalake-cpp-12.14.0 |  | h52c5a47_1      | 295 KB | conda-forge |
| backports.zstd-1.3.0                     |  | py311h6b1f9c4_0 | 239 KB | conda-forge |
| bokeh-3.9.0                              |  | pyhd8ed1ab_0    | 4.0 MB | conda-forge |
| brotli-1.2.0                             |  | hed03a55_1      | 20 KB  | conda-forge |
| brotli-bin-1.2.0                         |  | hb03c661_1      | 21 KB  | conda-forge |
| brotli-python-1.2.0                      |  | py311h66f275b_1 | 359 KB | conda-forge |
| bzip2-1.0.8                              |  | hda65f42_9      | 254 KB | conda-forge |
| c-ares-1.34.6                            |  | hb03c661_0      | 203 KB | conda-forge |
| ca-certificates-2026.2.25                |  | hbd8a1cb_0      | 144 KB | conda-forge |
| cairo-1.18.4                             |  | he90730b_1      | 966 KB | conda-forge |
| click-8.3.1                              |  | pyh8f84b5b_1    | 95 KB  | conda-forge |
| cloudpickle-3.1.2                        |  | pyhcf101f3_1    | 27 KB  | conda-forge |
| contourpy-1.3.3                          |  | py311h724c32c_4 | 313 KB | conda-forge |
| cycler-0.12.1                            |  | pyhcf101f3_2    | 14 KB  | conda-forge |
| cyrus-sasl-2.1.28                        |  | hac629b4_1      | 205 KB | conda-forge |
| cytoolz-1.1.0                            |  | py311h49ec1c0_2 | 609 KB | conda-forge |
| dask-2026.3.0                            |  | pyhc364b38_0    | 11 KB  | conda-forge |
| dask-core-2026.3.0                       |  | pyhc364b38_0    | 1.0 MB | conda-forge |
| dbus-1.16.2                              |  | h24cb091_1      | 437 KB | conda-forge |
| distributed-2026.3.0                     |  | pyhc364b38_0    | 826 KB | conda-forge |
| double-conversion-3.4.0                  |  | hecca717_0      | 70 KB  | conda-forge |
| font-ttf-dejavu-sans-mono-2.37           |  | hab24e00_0      | 388 KB | conda-forge |
| font-ttf-inconsolata-3.000               |  | h77eed37_0      | 94 KB  | conda-forge |
| font-ttf-source-code-pro-2.038           |  | h77eed37_0      | 684 KB | conda-forge |
| font-ttf-ubuntu-0.83                     |  | h77eed37_3      | 1.5 MB | conda-forge |
| fontconfig-2.17.1                        |  | h27c8c51_0      | 264 KB | conda-forge |
| fonts-conda-ecosystem-1                  |  | 0               | 4 KB   | conda-forge |
| fonts-conda-forge-1                      |  | hc364b38_1      | 4 KB   | conda-forge |
| fonttools-4.62.0                         |  | py311h3778330_0 | 2.9 MB | conda-forge |

|                           |  |                      |         |             |
|---------------------------|--|----------------------|---------|-------------|
| freetype-2.14.3           |  | ha770c72_0           | 170 KB  | conda-forge |
| fsspec-2026.2.0           |  | pyhd8ed1ab_0         | 145 KB  | conda-forge |
| gflags-2.2.2              |  | h5888daf_1005        | 117 KB  | conda-forge |
| glog-0.7.1                |  | hbabe93e_0           | 140 KB  | conda-forge |
| graphite2-1.3.14          |  | hecca717_2           | 97 KB   | conda-forge |
| h2-4.3.0                  |  | pyhcf101f3_0         | 94 KB   | conda-forge |
| harfbuzz-13.2.1           |  | h6083320_0           | 2.3 MB  | conda-forge |
| hpack-4.1.0               |  | pyhd8ed1ab_0         | 30 KB   | conda-forge |
| hyperframe-6.1.0          |  | pyhd8ed1ab_0         | 17 KB   | conda-forge |
| icu-78.3                  |  | h33c6efd_0           | 12.1 MB | conda-forge |
| importlib-metadata-8.8.0  |  | pyhcf101f3_0         | 34 KB   | conda-forge |
| jinja2-3.1.6              |  | pyhcf101f3_1         | 118 KB  | conda-forge |
| keyutils-1.6.3            |  | hb9d3cd8_0           | 131 KB  | conda-forge |
| kiwisolver-1.5.0          |  | py311h724c32c_0      | 76 KB   | conda-forge |
| krb5-1.22.2               |  | ha1258a1_0           | 1.3 MB  | conda-forge |
| lcms2-2.18                |  | h0c24ade_0           | 244 KB  | conda-forge |
| ld_impl_linux-64-2.45.1   |  | default_hbd61a6d_102 | 711 KB  | conda-forge |
| lerc-4.1.0                |  | hdb68285_0           | 255 KB  | conda-forge |
| libabseil-20260107.1      |  | cxx17_h7b12aa8_0     | 1.3 MB  | conda-forge |
| libarrow-23.0.1           |  | ha7f89c6_9_cpu       | 6.2 MB  | conda-forge |
| libarrow-acero-23.0.1     |  | h635bf11_9_cpu       | 595 KB  | conda-forge |
| libarrow-compute-23.0.1   |  | h53684a4_9_cpu       | 2.9 MB  | conda-forge |
| libarrow-dataset-23.0.1   |  | h635bf11_9_cpu       | 593 KB  | conda-forge |
| libarrow-substrait-23.0.1 |  | hb4dd7c2_9_cpu       | 507 KB  | conda-forge |
| libblas-3.11.0            |  | 6_h4a7cf45_openblas  | 18 KB   | conda-forge |
| libbrotlicommon-1.2.0     |  | hb03c661_1           | 78 KB   | conda-forge |
| libbrotlidec-1.2.0        |  | hb03c661_1           | 34 KB   | conda-forge |
| libbrotlienc-1.2.0        |  | hb03c661_1           | 291 KB  | conda-forge |
| libcblas-3.11.0           |  | 6_h0358290_openblas  | 18 KB   | conda-forge |
| libclang-cpp22.1-22.1.0   |  | default_h99862b1_0   | 20.7 MB | conda-forge |
| libclang13-22.1.0         |  | default_h746c552_0   | 12.2 MB | conda-forge |
| libcrc32c-1.1.2           |  | h9c3ff4c_0           | 20 KB   | conda-forge |
| libcups-2.3.3             |  | h7a8fb5f_6           | 4.3 MB  | conda-forge |
| libcurl-8.19.0            |  | hcf29cc6_0           | 456 KB  | conda-forge |
| libdeflate-1.25           |  | h17f619e_0           | 72 KB   | conda-forge |
| libdrm-2.4.125            |  | hb03c661_1           | 304 KB  | conda-forge |
| libedit-3.1.20250104      |  | p15321h7949ede_0     | 132 KB  | conda-forge |
| libegl-1.7.0              |  | ha4b6fd6_2           | 44 KB   | conda-forge |
| libev-4.33                |  | hd590300_2           | 110 KB  | conda-forge |
| libevent-2.1.12           |  | hf998b51_1           | 417 KB  | conda-forge |
| libexpat-2.7.4            |  | hecca717_0           | 75 KB   | conda-forge |
| libffi-3.5.2              |  | h3435931_0           | 57 KB   | conda-forge |
| libfreetype-2.14.3        |  | ha770c72_0           | 8 KB    | conda-forge |
| libfreetype6-2.14.3       |  | h73754d4_0           | 376 KB  | conda-forge |
| libgcc-15.2.0             |  | he0feb66_18          | 1017 KB | conda-forge |
| libgcc-ng-15.2.0          |  | h69a702a_18          | 27 KB   | conda-forge |
| libgfortran-15.2.0        |  | h69a702a_18          | 27 KB   | conda-forge |
| libgfortran5-15.2.0       |  | h68bc16d_18          | 2.4 MB  | conda-forge |

|                                     |  |                     |         |             |
|-------------------------------------|--|---------------------|---------|-------------|
| libgl-1.7.0                         |  | ha4b6fd6_2          | 132 KB  | conda-forge |
| libglib-2.86.4                      |  | h6548e54_1          | 4.2 MB  | conda-forge |
| libglvnd-1.7.0                      |  | ha4b6fd6_2          | 129 KB  | conda-forge |
| libglx-1.7.0                        |  | ha4b6fd6_2          | 74 KB   | conda-forge |
| libgomp-15.2.0                      |  | he0feb66_18         | 589 KB  | conda-forge |
| libgoogle-cloud-3.3.0               |  | h25dbb67_1          | 2.4 MB  | conda-forge |
| libgoogle-cloud-storage-3.3.0       |  | hdbdcf42_1          | 761 KB  | conda-forge |
| libgrpc-1.78.1                      |  | h1d1128b_0          | 6.7 MB  | conda-forge |
| libiconv-1.18                       |  | h3b78370_2          | 772 KB  | conda-forge |
| libjpeg-turbo-3.1.2                 |  | hb03c661_0          | 619 KB  | conda-forge |
| liblapack-3.11.0                    |  | 6_h47877c9_openblas | 18 KB   | conda-forge |
| libllvm22-22.1.2                    |  | hf7376ad_0          | 42.2 MB | conda-forge |
| liblzma-5.8.2                       |  | hb03c661_0          | 111 KB  | conda-forge |
| libnghttp2-1.68.1                   |  | h877daf1_0          | 648 KB  | conda-forge |
| libnsl-2.0.1                        |  | hb9d3cd8_1          | 33 KB   | conda-forge |
| libntlm-1.8                         |  | hb9d3cd8_0          | 33 KB   | conda-forge |
| libopenblas-0.3.32                  |  | pthread_h94d23a6_0  | 5.7 MB  | conda-forge |
| libopengl-1.7.0                     |  | ha4b6fd6_2          | 50 KB   | conda-forge |
| libopentelemetry-cpp-1.26.0         |  | h9692893_0          | 912 KB  | conda-forge |
| libopentelemetry-cpp-headers-1.26.0 |  | ha770c72_0          | 387 KB  | conda-forge |
| libparquet-23.0.1                   |  | h7376487_9_cpu      | 1.3 MB  | conda-forge |
| libpciaccess-0.18                   |  | hb9d3cd8_0          | 28 KB   | conda-forge |
| libpng-1.6.56                       |  | h421ea60_0          | 310 KB  | conda-forge |
| libpq-18.3                          |  | h9abb657_0          | 2.7 MB  | conda-forge |
| libprotobuf-6.33.5                  |  | h2b00c02_0          | 3.5 MB  | conda-forge |
| libre2-11-2025.11.05                |  | h0dc7533_1          | 208 KB  | conda-forge |
| libsqlite-3.52.0                    |  | hf4e2dac_0          | 929 KB  | conda-forge |
| libssh2-1.11.1                      |  | hcf80075_0          | 298 KB  | conda-forge |
| libstdcxx-15.2.0                    |  | h934dc35e_18        | 5.6 MB  | conda-forge |
| libstdcxx-ng-15.2.0                 |  | hdf11a46_18         | 27 KB   | conda-forge |
| libthrift-0.22.0                    |  | h454ac66_1          | 414 KB  | conda-forge |
| libtiff-4.7.1                       |  | h9d88235_1          | 425 KB  | conda-forge |
| libutf8proc-2.11.3                  |  | hfe17d71_0          | 84 KB   | conda-forge |
| libuuid-2.41.3                      |  | h5347b49_0          | 39 KB   | conda-forge |
| libvulkan-loader-1.4.341.0          |  | h5279c79_0          | 195 KB  | conda-forge |
| libwebp-base-1.6.0                  |  | hd42ef1d_0          | 419 KB  | conda-forge |
| libxcb-1.17.0                       |  | h8a09558_0          | 387 KB  | conda-forge |
| libxcrypt-4.4.36                    |  | hd590300_1          | 98 KB   | conda-forge |
| libxkbcommon-1.13.1                 |  | hca5e8e5_0          | 818 KB  | conda-forge |
| libxml2-2.15.2                      |  | he237659_0          | 45 KB   | conda-forge |
| libxml2-16-2.15.2                   |  | hca6bf5a_0          | 544 KB  | conda-forge |
| libxslt-1.1.43                      |  | h711ed8c_1          | 240 KB  | conda-forge |
| libzlib-1.3.2                       |  | h25fd6f3_2          | 62 KB   | conda-forge |
| llvmlite-0.46.0                     |  | py311h41a00d4_0     | 32.5 MB | conda-forge |
| locket-1.0.0                        |  | pyhd8ed1ab_0        | 8 KB    | conda-forge |
| lz4-4.4.5                           |  | py311h1c460e0_1     | 43 KB   | conda-forge |
| lz4-c-1.10.0                        |  | h5888daf_1          | 163 KB  | conda-forge |
| markupsafe-3.0.3                    |  | py311h3778330_1     | 26 KB   | conda-forge |

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| matplotlib-3.10.8           | py311h38be061_0     | 17 KB   | conda-forge |
| matplotlib-base-3.10.8      | py311h0f3be63_0     | 7.9 MB  | conda-forge |
| msgpack-python-1.1.2        | py311hdf67eae_1     | 101 KB  | conda-forge |
| munkres-1.1.4               | pyhd8ed1ab_1        | 15 KB   | conda-forge |
| narwhals-2.18.1             | pyhcf101f3_1        | 274 KB  | conda-forge |
| ncurses-6.5                 | h2d0b736_3          | 871 KB  | conda-forge |
| nlohmann_json-3.12.0        | h54a6638_1          | 133 KB  | conda-forge |
| numba-0.64.0                | py311h3c884d5_0     | 5.6 MB  | conda-forge |
| numpy-2.4.3                 | py311h2e04523_0     | 8.9 MB  | conda-forge |
| openjpeg-2.5.4              | h55fea9a_0          | 347 KB  | conda-forge |
| openldap-2.6.10             | hbde042b_1          | 767 KB  | conda-forge |
| openssl-3.6.1               | h35e630c_1          | 3.0 MB  | conda-forge |
| orc-2.3.0                   | h21090e2_0          | 1.4 MB  | conda-forge |
| packaging-26.0              | pyhcf101f3_0        | 70 KB   | conda-forge |
| pandas-3.0.1                | py311h8032f78_0     | 14.4 MB | conda-forge |
| partd-1.4.2                 | pyhd8ed1ab_0        | 20 KB   | conda-forge |
| pcre2-10.47                 | haa7fec5_0          | 1.2 MB  | conda-forge |
| pillow-12.1.1               | py311hf88fc01_0     | 1022 KB | conda-forge |
| pip-26.0.1                  | pyh8b19718_0        | 1.1 MB  | conda-forge |
| pixman-0.46.4               | h54a6638_1          | 440 KB  | conda-forge |
| prometheus-cpp-1.3.0        | ha5d0236_0          | 195 KB  | conda-forge |
| psutil-7.2.2                | py311haee01d2_0     | 226 KB  | conda-forge |
| pthread-stubs-0.4           | hb9d3cd8_1002       | 8 KB    | conda-forge |
| pyarrow-23.0.1              | py311h38be061_0     | 28 KB   | conda-forge |
| pyarrow-core-23.0.1         | py311h66caee6_0_cpu | 5.1 MB  | conda-forge |
| pyparsing-3.3.2             | pyhcf101f3_0        | 108 KB  | conda-forge |
| pyside6-6.10.2              | py311hf27b23e_1     | 12.5 MB | conda-forge |
| pysocks-1.7.1               | pyha55dd90_7        | 21 KB   | conda-forge |
| python-3.11.15              | hd63d673_0_cpython  | 29.5 MB | conda-forge |
| python-dateutil-2.9.0.post0 | pyhe01879c_2        | 228 KB  | conda-forge |
| python_abi-3.11             | 8_cp311             | 7 KB    | conda-forge |
| pyyaml-6.0.3                | py311h3778330_1     | 201 KB  | conda-forge |
| qhull-2020.2                | h434a139_5          | 540 KB  | conda-forge |
| qt6-main-6.10.2             | pl5321h16c4a6b_6    | 55.4 MB | conda-forge |
| re2-2025.11.05              | h5301d42_1          | 27 KB   | conda-forge |
| readline-8.3                | h853b02a_0          | 337 KB  | conda-forge |
| s2n-1.7.1                   | h1cbb8d7_1          | 386 KB  | conda-forge |
| setuptools-82.0.1           | pyh332efcf_0        | 625 KB  | conda-forge |
| six-1.17.0                  | pyhe01879c_1        | 18 KB   | conda-forge |
| snappy-1.2.2                | h03e3b7b_1          | 45 KB   | conda-forge |
| sortedcontainers-2.4.0      | pyhd8ed1ab_1        | 28 KB   | conda-forge |
| tblib-3.2.2                 | pyhcf101f3_0        | 19 KB   | conda-forge |
| tk-8.6.13                   | noxft_h366c992_103  | 3.1 MB  | conda-forge |
| toolz-1.1.0                 | pyhd8ed1ab_1        | 53 KB   | conda-forge |
| tornado-6.5.5               | py311h49ec1c0_0     | 856 KB  | conda-forge |
| tzdata-2025c                | hc9c84f9_1          | 116 KB  | conda-forge |
| unicodedata2-17.0.1         | py311h49ec1c0_0     | 400 KB  | conda-forge |
| urllib3-2.6.3               | pyhd8ed1ab_0        | 101 KB  | conda-forge |

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| wayland-1.25.0             |  | hd6090a7_0   | 326 KB   | conda-forge |
| wheel-0.46.3               |  | pyhd8ed1ab_0 | 31 KB    | conda-forge |
| xcb-util-0.4.1             |  | h4f16b4b_2   | 20 KB    | conda-forge |
| xcb-util-cursor-0.1.6      |  | hb03c661_0   | 20 KB    | conda-forge |
| xcb-util-image-0.4.0       |  | hb711507_2   | 24 KB    | conda-forge |
| xcb-util-keysyms-0.4.1     |  | hb711507_0   | 14 KB    | conda-forge |
| xcb-util-renderutil-0.3.10 |  | hb711507_0   | 17 KB    | conda-forge |
| xcb-util-wm-0.4.2          |  | hb711507_0   | 50 KB    | conda-forge |
| xkeyboard-config-2.47      |  | hb03c661_0   | 390 KB   | conda-forge |
| xorg-libice-1.1.2          |  | hb9d3cd8_0   | 57 KB    | conda-forge |
| xorg-libsm-1.2.6           |  | he73a12e_0   | 27 KB    | conda-forge |
| xorg-libx11-1.8.13         |  | he1eb515_0   | 820 KB   | conda-forge |
| xorg-libxau-1.0.12         |  | hb03c661_1   | 15 KB    | conda-forge |
| xorg-libxcomposite-0.4.7   |  | hb03c661_0   | 14 KB    | conda-forge |
| xorg-libxcursor-1.2.3      |  | hb9d3cd8_0   | 32 KB    | conda-forge |
| xorg-libxdamage-1.1.6      |  | hb9d3cd8_0   | 13 KB    | conda-forge |
| xorg-libxdmcp-1.1.5        |  | hb03c661_1   | 20 KB    | conda-forge |
| xorg-libxext-1.3.7         |  | hb03c661_0   | 49 KB    | conda-forge |
| xorg-libxfixes-6.0.2       |  | hb03c661_0   | 20 KB    | conda-forge |
| xorg-libxi-1.8.2           |  | hb9d3cd8_0   | 46 KB    | conda-forge |
| xorg-libxrandr-1.5.5       |  | hb03c661_0   | 30 KB    | conda-forge |
| xorg-libxrender-0.9.12     |  | hb9d3cd8_0   | 32 KB    | conda-forge |
| xorg-libxtst-1.2.5         |  | hb9d3cd8_3   | 32 KB    | conda-forge |
| xorg-libxxf86vm-1.1.7      |  | hb03c661_0   | 18 KB    | conda-forge |
| xyzservices-2025.11.0      |  | pyhd8ed1ab_0 | 50 KB    | conda-forge |
| yaml-0.2.5                 |  | h280c20c_3   | 83 KB    | conda-forge |
| zict-3.0.0                 |  | pyhd8ed1ab_1 | 35 KB    | conda-forge |
| zipp-3.23.0                |  | pyhcf101f3_1 | 24 KB    | conda-forge |
| zlib-1.3.2                 |  | h25fd6f3_2   | 94 KB    | conda-forge |
| zlib-ng-2.3.3              |  | hceb46e0_1   | 120 KB   | conda-forge |
| zstd-1.5.7                 |  | hb78ec9c_6   | 587 KB   | conda-forge |
| -----                      |  |              |          |             |
| Total:                     |  |              | 374.5 MB |             |

The following NEW packages will be INSTALLED:

|                    |                                                           |
|--------------------|-----------------------------------------------------------|
| _openmp_mutex      | conda-forge/linux-64::_openmp_mutex-4.5-20_gnu            |
| alsa-lib           | conda-forge/linux-64::alsa-lib-1.2.15.3-hb03c661_0        |
| aws-c-auth         | conda-forge/linux-64::aws-c-auth-0.10.1-h2d2dd48_2        |
| aws-c-cal          | conda-forge/linux-64::aws-c-cal-0.9.13-h2c9d079_1         |
| aws-c-common       | conda-forge/linux-64::aws-c-common-0.12.6-hb03c661_0      |
| aws-c-compression  | conda-forge/linux-64::aws-c-compression-0.3.2-h8b1a151_0  |
| aws-c-event-stream | conda-forge/linux-64::aws-c-event-stream-0.6.0-h9b893ba_1 |
| aws-c-http         | conda-forge/linux-64::aws-c-http-0.10.12-h4bacb7b_1       |
| aws-c-io           | conda-forge/linux-64::aws-c-io-0.26.3-hc87160b_0          |
| aws-c-mqtt         | conda-forge/linux-64::aws-c-mqtt-0.15.2-he9ea9c5_1        |
| aws-c-s3           | conda-forge/linux-64::aws-c-s3-0.11.5-h6d69fc9_5          |
| aws-c-sdkutils     | conda-forge/linux-64::aws-c-sdkutils-0.2.4-h8b1a151_4     |

|                    |                                                                           |
|--------------------|---------------------------------------------------------------------------|
| aws-checksums      | conda-forge/linux-64::aws-checksums-0.2.10-h8b1a151_0                     |
| aws-crt-cpp        | conda-forge/linux-64::aws-crt-cpp-0.37.4-h4c8aef7_3                       |
| aws-sdk-cpp        | conda-forge/linux-64::aws-sdk-cpp-1.11.747-hc3785e1_3                     |
| azure-core-cpp     | conda-forge/linux-64::azure-core-cpp-1.16.2-h206d751_0                    |
| azure-identity-cpp | conda-forge/linux-64::azure-identity-cpp-1.13.3-hed0cdb0_1                |
| azure-storage-blo~ | conda-forge/linux-64::azure-storage-blobs-cpp-12.16.0-hdd73cc9_1          |
| azure-storage-com~ | conda-forge/linux-64::azure-storage-common-cpp-12.12.0-ha7a2c86_1         |
| azure-storage-fil~ | conda-forge/linux-64::azure-storage-files-datalake-cpp-12.14.0-h52c5a47_1 |
| backports.zstd     | conda-forge/linux-64::backports.zstd-1.3.0-py311h6b1f9c4_0                |
| bokeh              | conda-forge/noarch::bokeh-3.9.0-pyhd8ed1ab_0                              |
| brotli             | conda-forge/linux-64::brotli-1.2.0-hed03a55_1                             |
| brotli-bin         | conda-forge/linux-64::brotli-bin-1.2.0-hb03c661_1                         |
| brotli-python      | conda-forge/linux-64::brotli-python-1.2.0-py311h66f275b_1                 |
| bzip2              | conda-forge/linux-64::bzip2-1.0.8-hda65f42_9                              |
| c-ares             | conda-forge/linux-64::c-ares-1.34.6-hb03c661_0                            |
| ca-certificates    | conda-forge/noarch::ca-certificates-2026.2.25-hbd8a1cb_0                  |
| cairo              | conda-forge/linux-64::cairo-1.18.4-he90730b_1                             |
| click              | conda-forge/noarch::click-8.3.1-pyh8f84b5b_1                              |
| cloudpickle        | conda-forge/noarch::cloudpickle-3.1.2-pyhcf101f3_1                        |
| contourpy          | conda-forge/linux-64::contourpy-1.3.3-py311h724c32c_4                     |
| cycler             | conda-forge/noarch::cycler-0.12.1-pyhcf101f3_2                            |
| cyrus-sasl         | conda-forge/linux-64::cyrus-sasl-2.1.28-hac629b4_1                        |
| cytoolz            | conda-forge/linux-64::cytoolz-1.1.0-py311h49ec1c0_2                       |
| dask               | conda-forge/noarch::dask-2026.3.0-pyhc364b38_0                            |
| dask-core          | conda-forge/noarch::dask-core-2026.3.0-pyhc364b38_0                       |
| dbus               | conda-forge/linux-64::dbus-1.16.2-h24cb091_1                              |
| distributed        | conda-forge/noarch::distributed-2026.3.0-pyhc364b38_0                     |
| double-conversion  | conda-forge/linux-64::double-conversion-3.4.0-hecca717_0                  |
| font-ttf-dejavu-s~ | conda-forge/noarch::font-ttf-dejavu-sans-mono-2.37-hab24e00_0             |
| font-ttf-inconsol~ | conda-forge/noarch::font-ttf-inconsolata-3.000-h77eed37_0                 |
| font-ttf-source-c~ | conda-forge/noarch::font-ttf-source-code-pro-2.038-h77eed37_0             |
| font-ttf-ubuntu    | conda-forge/noarch::font-ttf-ubuntu-0.83-h77eed37_3                       |
| fontconfig         | conda-forge/linux-64::fontconfig-2.17.1-h27c8c51_0                        |
| fonts-conda-ecosy~ | conda-forge/noarch::fonts-conda-ecosystem-1-0                             |
| fonts-conda-forge  | conda-forge/noarch::fonts-conda-forge-1-hc364b38_1                        |
| fonttools          | conda-forge/linux-64::fonttools-4.62.0-py311h3778330_0                    |
| freetype           | conda-forge/linux-64::freetype-2.14.3-ha770c72_0                          |
| fsspec             | conda-forge/noarch::fsspec-2026.2.0-pyhd8ed1ab_0                          |
| gflags             | conda-forge/linux-64::gflags-2.2.2-h5888daf_1005                          |
| glog               | conda-forge/linux-64::glog-0.7.1-hbabe93e_0                               |
| graphite2          | conda-forge/linux-64::graphite2-1.3.14-hecca717_2                         |
| h2                 | conda-forge/noarch::h2-4.3.0-pyhcf101f3_0                                 |
| harfbuzz           | conda-forge/linux-64::harfbuzz-13.2.1-h6083320_0                          |
| hpack              | conda-forge/noarch::hpack-4.1.0-pyhd8ed1ab_0                              |
| hyperframe         | conda-forge/noarch::hyperframe-6.1.0-pyhd8ed1ab_0                         |
| icu                | conda-forge/linux-64::icu-78.3-h33c6efd_0                                 |
| importlib-metadata | conda-forge/noarch::importlib-metadata-8.8.0-pyhcf101f3_0                 |
| jinja2             | conda-forge/noarch::jinja2-3.1.6-pyhcf101f3_1                             |

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| keyutils           | conda-forge/linux-64::keyutils-1.6.3-hb9d3cd8_0                    |
| kiwisolver         | conda-forge/linux-64::kiwisolver-1.5.0-py311h724c32c_0             |
| krb5               | conda-forge/linux-64::krb5-1.22.2-ha1258a1_0                       |
| lcms2              | conda-forge/linux-64::lcms2-2.18-h0c24ade_0                        |
| ld_impl_linux-64   | conda-forge/linux-64::ld_impl_linux-64-2.45.1-default_hbd61a6d_102 |
| lerc               | conda-forge/linux-64::lerc-4.1.0-hdb68285_0                        |
| libabseil          | conda-forge/linux-64::libabseil-20260107.1-cxx17_h7b12aa8_0        |
| libarrow           | conda-forge/linux-64::libarrow-23.0.1-ha7f89c6_9_cpu               |
| libarrow-acero     | conda-forge/linux-64::libarrow-acero-23.0.1-h635bf11_9_cpu         |
| libarrow-compute   | conda-forge/linux-64::libarrow-compute-23.0.1-h53684a4_9_cpu       |
| libarrow-dataset   | conda-forge/linux-64::libarrow-dataset-23.0.1-h635bf11_9_cpu       |
| libarrow-substrait | conda-forge/linux-64::libarrow-substrait-23.0.1-hb4dd7c2_9_cpu     |
| libblas            | conda-forge/linux-64::libblas-3.11.0-6_h4a7cf45_openblas           |
| libbrotlicommon    | conda-forge/linux-64::libbrotlicommon-1.2.0-hb03c661_1             |
| libbrotlidec       | conda-forge/linux-64::libbrotlidec-1.2.0-hb03c661_1                |
| libbrotlienc       | conda-forge/linux-64::libbrotlienc-1.2.0-hb03c661_1                |
| libcbblas          | conda-forge/linux-64::libcbblas-3.11.0-6_h0358290_openblas         |
| libclang-cpp22.1   | conda-forge/linux-64::libclang-cpp22.1-22.1.0-default_h99862b1_0   |
| libclang13         | conda-forge/linux-64::libclang13-22.1.0-default_h746c552_0         |
| libcrc32c          | conda-forge/linux-64::libcrc32c-1.1.2-h9c3ff4c_0                   |
| libcups            | conda-forge/linux-64::libcups-2.3.3-h7a8fb5f_6                     |
| libcurl            | conda-forge/linux-64::libcurl-8.19.0-hcf29cc6_0                    |
| libdeflate         | conda-forge/linux-64::libdeflate-1.25-h17f619e_0                   |
| libdrm             | conda-forge/linux-64::libdrm-2.4.125-hb03c661_1                    |
| libedit            | conda-forge/linux-64::libedit-3.1.20250104-pl5321h7949ede_0        |
| libegl             | conda-forge/linux-64::libegl-1.7.0-ha4b6fd6_2                      |
| libev              | conda-forge/linux-64::libev-4.33-hd590300_2                        |
| libevent           | conda-forge/linux-64::libevent-2.1.12-hf998b51_1                   |
| libexpat           | conda-forge/linux-64::libexpat-2.7.4-hecca717_0                    |
| libffi             | conda-forge/linux-64::libffi-3.5.2-h3435931_0                      |
| libfreetype        | conda-forge/linux-64::libfreetype-2.14.3-ha770c72_0                |
| libfreetype6       | conda-forge/linux-64::libfreetype6-2.14.3-h73754d4_0               |
| libgcc             | conda-forge/linux-64::libgcc-15.2.0-he0feb66_18                    |
| libgcc-ng          | conda-forge/linux-64::libgcc-ng-15.2.0-h69a702a_18                 |
| libgfortran        | conda-forge/linux-64::libgfortran-15.2.0-h69a702a_18               |
| libgfortran5       | conda-forge/linux-64::libgfortran5-15.2.0-h68bc16d_18              |
| libgl              | conda-forge/linux-64::libgl-1.7.0-ha4b6fd6_2                       |
| libglib            | conda-forge/linux-64::libglib-2.86.4-h6548e54_1                    |
| libglvnd           | conda-forge/linux-64::libglvnd-1.7.0-ha4b6fd6_2                    |
| libglx             | conda-forge/linux-64::libglx-1.7.0-ha4b6fd6_2                      |
| libgomp            | conda-forge/linux-64::libgomp-15.2.0-he0feb66_18                   |
| libgoogle-cloud    | conda-forge/linux-64::libgoogle-cloud-3.3.0-h25dbb67_1             |
| libgoogle-cloud-s~ | conda-forge/linux-64::libgoogle-cloud-storage-3.3.0-hdbdcf42_1     |
| libgrpc            | conda-forge/linux-64::libgrpc-1.78.1-h1d1128b_0                    |
| libiconv           | conda-forge/linux-64::libiconv-1.18-h3b78370_2                     |
| libjpeg-turbo      | conda-forge/linux-64::libjpeg-turbo-3.1.2-hb03c661_0               |
| liblapack          | conda-forge/linux-64::liblapack-3.11.0-6_h47877c9_openblas         |
| libllvm22          | conda-forge/linux-64::libllvm22-22.1.2-hf7376ad_0                  |

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|-------------------|----------------------------------------------------------------------|
| liblzma           | conda-forge/linux-64::liblzma-5.8.2-hb03c661_0                       |
| libnghttp2        | conda-forge/linux-64::libnghttp2-1.68.1-h877daf1_0                   |
| libnsl            | conda-forge/linux-64::libnsl-2.0.1-hb9d3cd8_1                        |
| libntlm           | conda-forge/linux-64::libntlm-1.8-hb9d3cd8_0                         |
| libopenblas       | conda-forge/linux-64::libopenblas-0.3.32-pthreads_h94d23a6_0         |
| libopengl         | conda-forge/linux-64::libopengl-1.7.0-ha4b6fd6_2                     |
| libopentelemetry~ | conda-forge/linux-64::libopentelemetry-cpp-1.26.0-h9692893_0         |
| libopentelemetry~ | conda-forge/linux-64::libopentelemetry-cpp-headers-1.26.0-ha770c72_0 |
| libparquet        | conda-forge/linux-64::libparquet-23.0.1-h7376487_9_cpu               |
| libpciaccess      | conda-forge/linux-64::libpciaccess-0.18-hb9d3cd8_0                   |
| libpng            | conda-forge/linux-64::libpng-1.6.56-h421ea60_0                       |
| libpq             | conda-forge/linux-64::libpq-18.3-h9abb657_0                          |
| libprotobuf       | conda-forge/linux-64::libprotobuf-6.33.5-h2b00c02_0                  |
| libre2-11         | conda-forge/linux-64::libre2-11-2025.11.05-h0dc7533_1                |
| libsqlite         | conda-forge/linux-64::libsqlite-3.52.0-hf4e2dac_0                    |
| libssh2           | conda-forge/linux-64::libssh2-1.11.1-hcf80075_0                      |
| libstdcxx         | conda-forge/linux-64::libstdcxx-15.2.0-h934c35e_18                   |
| libstdcxx-ng      | conda-forge/linux-64::libstdcxx-ng-15.2.0-hdf11a46_18                |
| libthrift         | conda-forge/linux-64::libthrift-0.22.0-h454ac66_1                    |
| libtiff           | conda-forge/linux-64::libtiff-4.7.1-h9d88235_1                       |
| libutf8proc       | conda-forge/linux-64::libutf8proc-2.11.3-hfe17d71_0                  |
| libuuid           | conda-forge/linux-64::libuuid-2.41.3-h5347b49_0                      |
| libvulkan-loader  | conda-forge/linux-64::libvulkan-loader-1.4.341.0-h5279c79_0          |
| libwebp-base      | conda-forge/linux-64::libwebp-base-1.6.0-hd42ef1d_0                  |
| libxcb            | conda-forge/linux-64::libxcb-1.17.0-h8a09558_0                       |
| libxcrypt         | conda-forge/linux-64::libxcrypt-4.4.36-hd590300_1                    |
| libxkbcommon      | conda-forge/linux-64::libxkbcommon-1.13.1-hca5e8e5_0                 |
| libxml2           | conda-forge/linux-64::libxml2-2.15.2-he237659_0                      |
| libxml2-16        | conda-forge/linux-64::libxml2-16-2.15.2-hca6bf5a_0                   |
| libxslt           | conda-forge/linux-64::libxslt-1.1.43-h711ed8c_1                      |
| libzlib           | conda-forge/linux-64::libzlib-1.3.2-h25fd6f3_2                       |
| llvmlite          | conda-forge/linux-64::llvmlite-0.46.0-py311h41a00d4_0                |
| loket             | conda-forge/noarch::loket-1.0.0-pyhd8ed1ab_0                         |
| lz4               | conda-forge/linux-64::lz4-4.4.5-py311h1c460e0_1                      |
| lz4-c             | conda-forge/linux-64::lz4-c-1.10.0-h5888daf_1                        |
| markupsafe        | conda-forge/linux-64::markupsafe-3.0.3-py311h3778330_1               |
| matplotlib        | conda-forge/linux-64::matplotlib-3.10.8-py311h38be061_0              |
| matplotlib-base   | conda-forge/linux-64::matplotlib-base-3.10.8-py311h0f3be63_0         |
| msgpack-python    | conda-forge/linux-64::msgpack-python-1.1.2-py311hdf67eae_1           |
| munkres           | conda-forge/noarch::munkres-1.1.4-pyhd8ed1ab_1                       |
| narwhals          | conda-forge/noarch::narwhals-2.18.1-pyhcf101f3_1                     |
| ncurses           | conda-forge/linux-64::ncurses-6.5-h2d0b736_3                         |
| nlohmann_json     | conda-forge/linux-64::nlohmann_json-3.12.0-h54a6638_1                |
| numba             | conda-forge/linux-64::numba-0.64.0-py311h3c884d5_0                   |
| numpy             | conda-forge/linux-64::numpy-2.4.3-py311h2e04523_0                    |
| openjpeg          | conda-forge/linux-64::openjpeg-2.5.4-h55fea9a_0                      |
| openldap          | conda-forge/linux-64::openldap-2.6.10-hbde042b_1                     |
| openssl           | conda-forge/linux-64::openssl-3.6.1-h35e630c_1                       |

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| orc                | conda-forge/linux-64::orc-2.3.0-h21090e2_0                    |
| packaging          | conda-forge/noarch::packaging-26.0-pyhcf101f3_0               |
| pandas             | conda-forge/linux-64::pandas-3.0.1-py311h8032f78_0            |
| partd              | conda-forge/noarch::partd-1.4.2-pyhd8ed1ab_0                  |
| pcre2              | conda-forge/linux-64::pcre2-10.47-haa7fec5_0                  |
| pillow             | conda-forge/linux-64::pillow-12.1.1-py311hf88fc01_0           |
| pip                | conda-forge/noarch::pip-26.0.1-pyh8b19718_0                   |
| pixman             | conda-forge/linux-64::pixman-0.46.4-h54a6638_1                |
| prometheus-cpp     | conda-forge/linux-64::prometheus-cpp-1.3.0-ha5d0236_0         |
| psutil             | conda-forge/linux-64::psutil-7.2.2-py311haee01d2_0            |
| pthread-stubs      | conda-forge/linux-64::pthread-stubs-0.4-hb9d3cd8_1002         |
| pyarrow            | conda-forge/linux-64::pyarrow-23.0.1-py311h38be061_0          |
| pyarrow-core       | conda-forge/linux-64::pyarrow-core-23.0.1-py311h66caee6_0_cpu |
| pyparsing          | conda-forge/noarch::pyparsing-3.3.2-pyhcf101f3_0              |
| pyside6            | conda-forge/linux-64::pyside6-6.10.2-py311hf27b23e_1          |
| pysocks            | conda-forge/noarch::pysocks-1.7.1-pyha55dd90_7                |
| python             | conda-forge/linux-64::python-3.11.15-hd63d673_0_cpython       |
| python-dateutil    | conda-forge/noarch::python-dateutil-2.9.0.post0-pyhe01879c_2  |
| python_abi         | conda-forge/noarch::python_abi-3.11-8_cp311                   |
| pyyaml             | conda-forge/linux-64::pyyaml-6.0.3-py311h3778330_1            |
| qhull              | conda-forge/linux-64::qhull-2020.2-h434a139_5                 |
| qt6-main           | conda-forge/linux-64::qt6-main-6.10.2-pl5321h16c4a6b_6        |
| re2                | conda-forge/linux-64::re2-2025.11.05-h5301d42_1               |
| readline           | conda-forge/linux-64::readline-8.3-h853b02a_0                 |
| s2n                | conda-forge/linux-64::s2n-1.7.1-h1cbb8d7_1                    |
| setuptools         | conda-forge/noarch::setuptools-82.0.1-pyh332efcf_0            |
| six                | conda-forge/noarch::six-1.17.0-pyhe01879c_1                   |
| snappy             | conda-forge/linux-64::snappy-1.2.2-h03e3b7b_1                 |
| sortedcontainers   | conda-forge/noarch::sortedcontainers-2.4.0-pyhd8ed1ab_1       |
| tblib              | conda-forge/noarch::tblib-3.2.2-pyhcf101f3_0                  |
| tk                 | conda-forge/linux-64::tk-8.6.13-noxft_h366c992_103            |
| toolz              | conda-forge/noarch::toolz-1.1.0-pyhd8ed1ab_1                  |
| tornado            | conda-forge/linux-64::tornado-6.5.5-py311h49ec1c0_0           |
| tzdata             | conda-forge/noarch::tzdata-2025c-hc9c84f9_1                   |
| unicodedata2       | conda-forge/linux-64::unicodedata2-17.0.1-py311h49ec1c0_0     |
| urllib3            | conda-forge/noarch::urllib3-2.6.3-pyhd8ed1ab_0                |
| wayland            | conda-forge/linux-64::wayland-1.25.0-hd6090a7_0               |
| wheel              | conda-forge/noarch::wheel-0.46.3-pyhd8ed1ab_0                 |
| xcb-util           | conda-forge/linux-64::xcb-util-0.4.1-h4f16b4b_2               |
| xcb-util-cursor    | conda-forge/linux-64::xcb-util-cursor-0.1.6-hb03c661_0        |
| xcb-util-image     | conda-forge/linux-64::xcb-util-image-0.4.0-hb711507_2         |
| xcb-util-keysyms   | conda-forge/linux-64::xcb-util-keysyms-0.4.1-hb711507_0       |
| xcb-util-renderut~ | conda-forge/linux-64::xcb-util-renderutil-0.3.10-hb711507_0   |
| xcb-util-wm        | conda-forge/linux-64::xcb-util-wm-0.4.2-hb711507_0            |
| xkeyboard-config   | conda-forge/linux-64::xkeyboard-config-2.47-hb03c661_0        |
| xorg-libice        | conda-forge/linux-64::xorg-libice-1.1.2-hb9d3cd8_0            |
| xorg-libsm         | conda-forge/linux-64::xorg-libsm-1.2.6-he73a12e_0             |
| xorg-libx11        | conda-forge/linux-64::xorg-libx11-1.8.13-he1eb515_0           |

```

xorg-libxau          conda-forge/linux-64::xorg-libxau-1.0.12-hb03c661_1
xorg-libxcomposite   conda-forge/linux-64::xorg-libxcomposite-0.4.7-hb03c661_0
xorg-libxcursor      conda-forge/linux-64::xorg-libxcursor-1.2.3-hb9d3cd8_0
xorg-libxdamage      conda-forge/linux-64::xorg-libxdamage-1.1.6-hb9d3cd8_0
xorg-libxdmcp        conda-forge/linux-64::xorg-libxdmcp-1.1.5-hb03c661_1
xorg-libxext         conda-forge/linux-64::xorg-libxext-1.3.7-hb03c661_0
xorg-libxfixes       conda-forge/linux-64::xorg-libxfixes-6.0.2-hb03c661_0
xorg-libxi           conda-forge/linux-64::xorg-libxi-1.8.2-hb9d3cd8_0
xorg-libxrandr       conda-forge/linux-64::xorg-libxrandr-1.5.5-hb03c661_0
xorg-libxrender      conda-forge/linux-64::xorg-libxrender-0.9.12-hb9d3cd8_0
xorg-libxtst         conda-forge/linux-64::xorg-libxtst-1.2.5-hb9d3cd8_3
xorg-libxxf86vm      conda-forge/linux-64::xorg-libxxf86vm-1.1.7-hb03c661_0
xyzservices          conda-forge/noarch::xyzservices-2025.11.0-pyhd8ed1ab_0
yaml                 conda-forge/linux-64::yaml-0.2.5-h280c20c_3
zict                 conda-forge/noarch::zict-3.0.0-pyhd8ed1ab_1
zipp                 conda-forge/noarch::zipp-3.23.0-pyhcf101f3_1
zlib                 conda-forge/linux-64::zlib-1.3.2-h25fd6f3_2
zlib-ng              conda-forge/linux-64::zlib-ng-2.3.3-hceb46e0_1
zstd                 conda-forge/linux-64::zstd-1.5.7-hb78ec9c_6

```

Downloading and Extracting Packages:

```

Preparing transaction: done
Verifying transaction: done
Executing transaction: done

```

```
#
```

```
# To activate this environment, use
```

```
#
```

```
#     $ conda activate nsc
```

```
#
```

```
# To deactivate an active environment, use
```

```
#
```

```
#     $ conda deactivate
```

See if it works.

```
(base) ubuntu@ncs00:~$ conda activate nsc
```

```
(nsc) ubuntu@ncs00:~$ python -c "import dask, numba, numpy; print(dask.__version__, numba.__version__, numpy.__version__)"
2026.3.0 0.64.0 2.4.3
```

### 1.1.6 1.6 Adapt your L06 code for the cluster

Three small changes are needed before your code will work on the cluster:

1. **\*\*\*\*Replace LocalCluster + Client\*\*** with a single Client line pointing at the scheduler:\*\*

```
# Remove these two lines:
# cluster = LocalCluster(n_workers=8, threads_per_worker=1)
# client = Client(cluster)
```

```
# Replace with:
client = Client("tcp://10.92.y.x:8786")
```

A plain `Client()` with no argument starts a new local scheduler instead of connecting to the cluster.

```
[1]: import numpy as np
from numba import njit
# From lecture 4

@njit(cache=True)
def mandelbrot_pixel(c_real, c_imag, max_iter):
    z_real = z_imag = 0.0
    for i in range(max_iter):
        zr2 = z_real*z_real
        zi2 = z_imag*z_imag
        if zr2 + zi2 > 4.0: return i
        z_imag = 2.0*z_real*z_imag + c_imag
        z_real = zr2 - zi2 + c_real
    return max_iter

@njit(cache=True)
def mandelbrot_chunk(row_start, row_end, N, x_min, x_max, y_min, y_max,
    ↪max_iter):
    out = np.empty((row_end - row_start, N), dtype=np.int32)
    dx = (x_max - x_min) / N
    dy = (y_max - y_min) / N
    for r in range(row_end - row_start):
        c_imag = y_min + (r + row_start) * dy
        for col in range(N):
            out[r, col] = mandelbrot_pixel(x_min + col*dx, c_imag, max_iter)
    return out
```

```
[3]: from dask import delayed
import dask
# From lecture 6

def mandelbrot_dask(N, x_min, x_max, y_min, y_max,
    max_iter=100, n_chunks=32):
    chunk_size = max(1, N // n_chunks)
    tasks, row = [], 0
    while row < N:
        row_end = min(row + chunk_size, N)
        tasks.append(delayed(mandelbrot_chunk)(
```

```

        row, row_end, N, x_min, x_max, y_min, y_max, max_iter))
    row = row_end
    parts = dask.compute(*tasks)
    return np.vstack(parts)

```

```

[ ]: import statistics
import time
from dask.distributed import LocalCluster
from dask.distributed import Client

# From lecture 6

N, max_iter = 1024, 100
X_MIN, X_MAX, Y_MIN, Y_MAX = -2.5, 1.0, -1.25, 1.25
client = Client("tcp://10.92.1.106:8786") # Use remote cluster at strato
client.run(lambda: mandelbrot_chunk(0, 8, 8, X_MIN, X_MAX, # warm up all
    ↪workers
                                Y_MIN, Y_MAX, 10))

n_chunks_arr = [1, 2, 4, 8, 16, 32]
times_res = []
lifs = []
vs1x = []
speedups = []

for n_chunks in n_chunks_arr:
    times = []
    for _ in range(3):
        t0 = time.perf_counter()
        result = mandelbrot_dask(
            N, X_MIN, X_MAX, Y_MIN, Y_MAX, max_iter, n_chunks)
        times.append(time.perf_counter() - t0)

    T_p = statistics.median(times)
    times_res.append(T_p)
    T_1 = times_res[0]
    vs1x.append(T_p / T_1)
    speedups.append(T_1 / T_p)
    lifs.append(8 * T_p / T_1 - 1)

client.close()

```

2. **\*\*\*\*Remove cluster.close()\*\*** — you do not own the cluster, only the client:\*\*

```

client.close()    # keep this
# cluster.close() # remove this

```

Already done. Please see the code above.

**1.1.7 3. \*\*\*\*Warm up workers\*\* before timing. If your L06 code only warmed up locally, add a `client.run` call:\*\***

```
client.run(lambda: your_numba_function(warmup_args))
```

This ensures Numba is JIT-compiled on each worker before the benchmark starts.

I already included “warmup” (compile time)

**Make these changes on your laptop, then copy the file to the VM.**

\*\*\*\*Option A — git clone (recommended if your code is on GitHub/GitLab):\*\*\*\*

Commit and push the changes, then on the VM:

```
git clone https://github.com/YOUR_USERNAME/YOUR_REPO.git
```

\*\*\*\*Option B — copy from your laptop:\*\*\*\*

```
scp -i ~/.ssh/my_ssh_key mandelbrot_dask.py ubuntu@10.92.y.x:~/
```

I have chosen to go with option B, but first I need to export this notebook to a script.

```
[6]: !jupyter nbconvert --to script 107_strato_dask_setup.ipynb --output_
    ↪mandelbrot_dask
```

```
[NbConvertApp] Converting notebook 107_strato_dask_setup.ipynb to script
```

```
[NbConvertApp] Writing 59853 bytes to mandelbrot_dask.py
```

```
(mikkel@pella)-[~/Sync/AAU/Semester 8/github/nsc-mikkel]
```

```
$ scp -i nsc mandelbrot_dask.py ubuntu@10.92.1.106:~/
```

```
mandelbrot_dask.py                                100%   59KB 123.3KB/s   00:00
```

### 1.1.8 1.7 Test locally

Run a quick sanity check on the VM to confirm Numba and Dask work before snapshotting:

```
conda activate nsc
python -c "
from dask.distributed import LocalCluster, Client
import dask
cluster = LocalCluster(n_workers=2, threads_per_worker=1)
client = Client(cluster)
result = dask.delayed(sum)(range(1000))
print(dask.compute(result))
client.close(); cluster.close()
"
```

Expected output (some startup log lines may appear before it):

```
(499500,)
```

If you see this, Dask is working correctly. Ignore any warnings about dashboard not being available or deprecated APIs.

I get the following output:

```
(base) ubuntu@nsc00:~$ conda activate nsc
python -c "
from dask.distributed import LocalCluster, Client
import dask
cluster = LocalCluster(n_workers=2, threads_per_worker=1)
client = Client(cluster)
result = dask.delayed(sum)(range(1000))
print(dask.compute(result))
client.close(); cluster.close()
"
(499500,)
```

It seems that i am good to go.

## 1.2 Part 2 — Snapshot and Launch Worker VMs

### 1.2.1 2.1 Create a snapshot

Shut down the VM before snapshotting to ensure a consistent disk image. From the terminal on the VM:

```
sudo shutdown -h now
```

Your SSH connection will drop — that is expected. Wait 30 seconds, then proceed in the Strato dashboard.

\*\*\*\*Do the snapshot from the Strato dashboard, not the terminal.\*\*\*\*

1. **Go to** Compute → Instances.
2. **Confirm the VM status is** Shutoff.
3. **Find your VM, click the dropdown arrow on the right, choose** Create Snapshot.
4. **Name it something like** nsc-dask-node **and click** Create Snapshot.

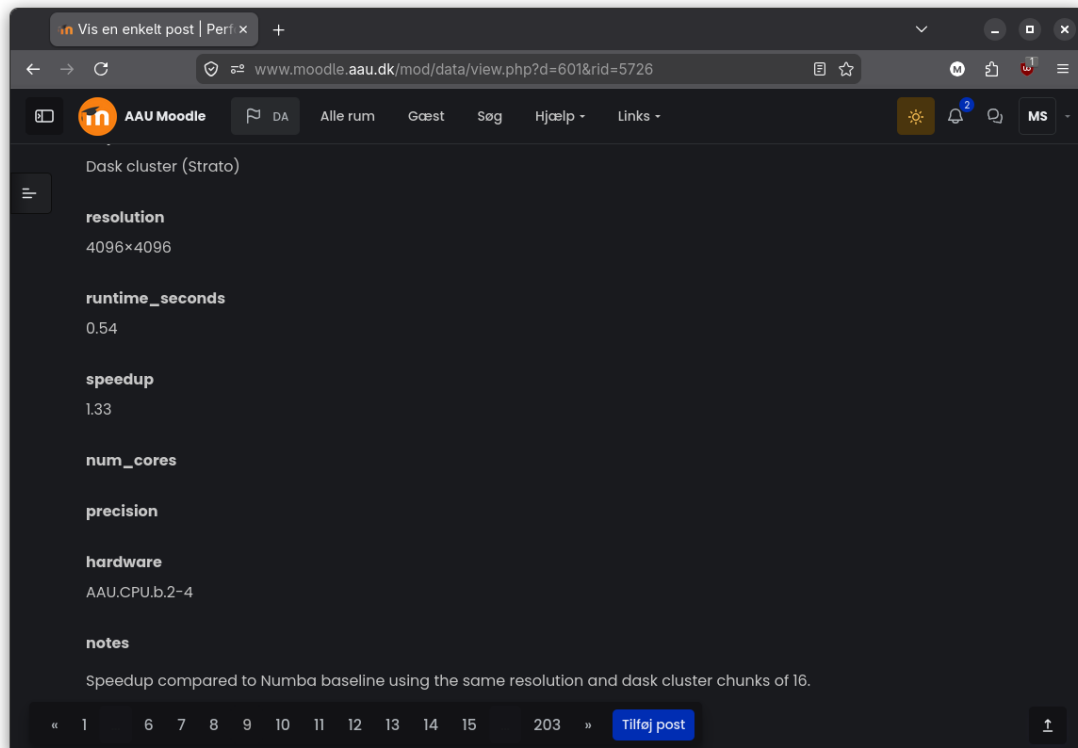
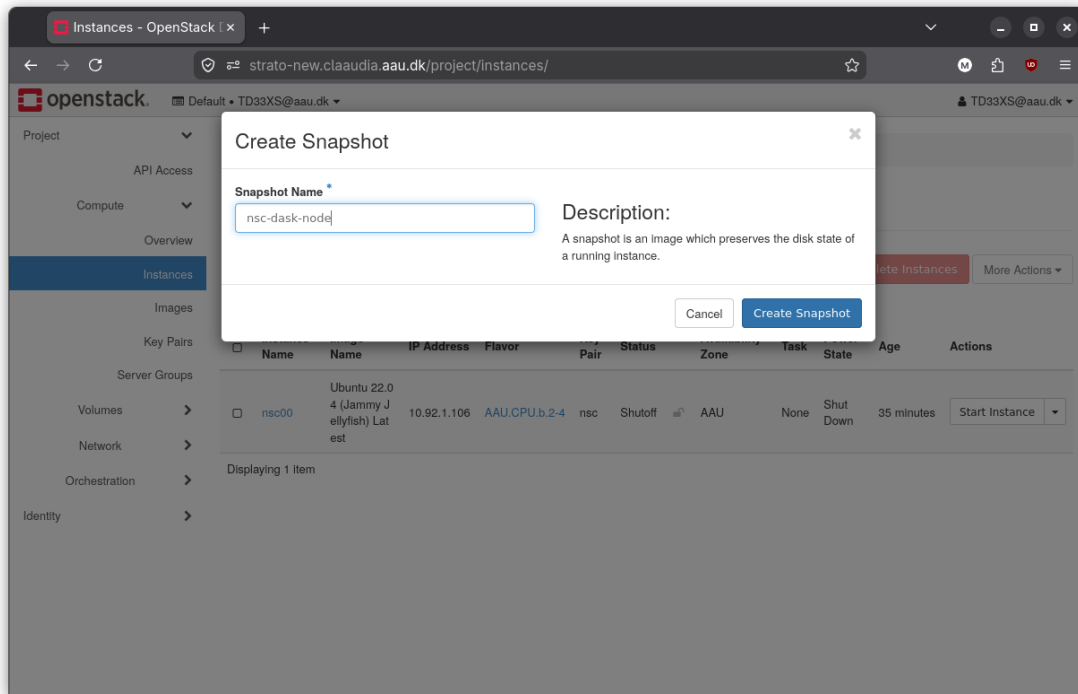
**Wait for the snapshot to finish (status changes to Active\*\* under Compute → Images).** This takes 1–3 minutes.\*\*

The snapshot captures the entire disk — Miniconda, your conda environment, and your code — so all worker VMs will be byte-for-byte identical to the head node. This prevents version mismatch errors.

I power off the vm.

```
(nsc) ubuntu@nsc00:~$ Connection to 10.92.1.106 closed by remote host.
Connection to 10.92.1.106 closed.
```

And create the snapshot.



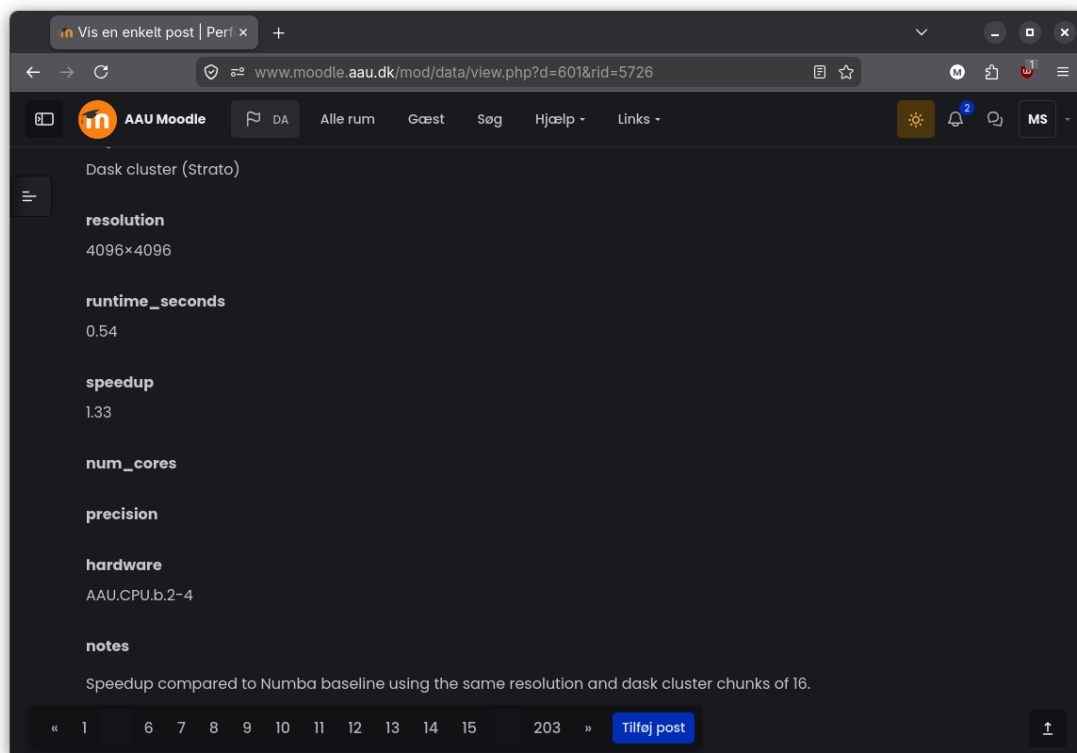
### 1.2.2 2.2 Launch worker VMs from the snapshot

1. Go to Compute → Images, find your **nsc-dask-node** snapshot.

2. Click **Launch** (not the image in the public list — your own snapshot under **Project**).
3. **Configure each worker VM:**
  - **Instance Name:** `nsc-worker-1` (repeat for `nsc-worker-2`, etc.)
  - **Source:** your snapshot (already selected)
    - **Flavor:** `AAU.CPU.b.2-4` (2 vCPUs, 4 GB RAM) — same as head node; launch 2–3 worker VMs for 4–6 total workers. Use `AAU.CPU.d.4-12` if you want 4 workers per VM (2 VMs = 8 workers total)
  - **Network:** same as head node
  - **Security Group:** same group (ports 22, 8786, 8787 already open)
  - **Key Pair:** same key
  - **No floating IP needed** on worker VMs
4. **Launch. Repeat to create 2–3 worker VMs.**

**Note the internal IP** of each worker from the Instances list — you will need them shortly.

**Before continuing**, start the head node again: **go to** Compute → Instances, **find the head node (status: Shutoff)**, **click the dropdown** → Start Instance. **Wait until status is Active**, then SSH back in.



## 1.3 Part 3 — Start the Dask Cluster

### 1.3.1 3.1 Note the head node's IP

**In the Strato dashboard under Compute → Instances, note the IP of your head node (e.g. 10.92.y.x). This is the scheduler address** that workers will connect to. Use this same IP in all subsequent steps.

We can see from the previous exercise that the IPv4 of the head node is 10.92.1.106

### 1.3.2 3.2 Start the scheduler on the head node

SSH into the head node. Use tmux so the scheduler keeps running even if your SSH connection drops:

```
tmux new -s dask
conda activate nsc
dask scheduler
```

The scheduler prints several log lines; look for:

```
Scheduler at:  tcp://10.92.y.x:8786
dashboard at:  http://10.92.y.x:8787/status
```

```
****Expected warnings — ignore these:** - FutureWarning: dask-scheduler is deprecated
— harmless, the command still works. - To route to workers diagnostics ... install
jupyter-server-proxy — not needed for our use.**
```

To detach from tmux without stopping the scheduler: press Ctrl-B, then D. To reattach later: `tmux attach -t dask`.

I get the following output.

```
(base) ubuntu@nsc00:~$ conda activate nsc
(nsc) ubuntu@nsc00:~$ dask scheduler
2026-03-27 21:13:39,368 - distributed.scheduler - INFO - -----
2026-03-27 21:13:42,807 - distributed.http.proxy - INFO - To route to workers diagnostics web s
2026-03-27 21:13:42,848 - distributed.scheduler - INFO - State start
2026-03-27 21:13:42,850 - distributed.scheduler - INFO - -----
2026-03-27 21:13:42,851 - distributed.scheduler - INFO - Scheduler at:  tcp://10.92.1.106:8
2026-03-27 21:13:42,852 - distributed.scheduler - INFO - dashboard at:  http://10.92.1.106:8
2026-03-27 21:13:42,852 - distributed.scheduler - INFO - Registering Worker plugin shuffle
```

### 1.3.3 3.3 Start a worker on a VM

These steps are the same for every worker VM.

Open a terminal on your laptop and SSH into the worker VM (e.g. 10.92.y.x):

```
ssh -i ~/.ssh/my_ssh_key ubuntu@10.92.y.x
```

Then on the VM:

```
conda activate nsc
dask worker 10.92.y.x:8786 --nworkers -1 --nthreads 1
```

Replace 10.92.y.x with your head node's IP\*\* (from step 3.1).\*\*

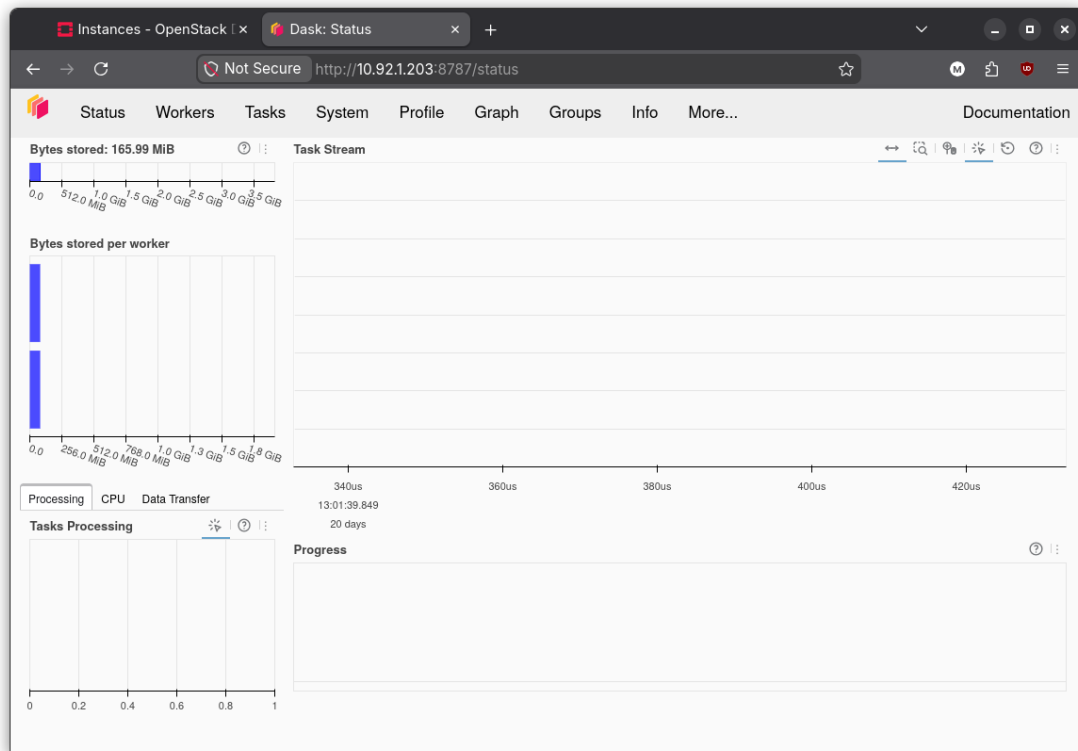
--nworkers -1 starts one worker process per available vCPU — so a AAU.CPU.b.2-4 VM contributes 2 worker processes. --nthreads 1 gives each process a single thread, which is correct for Numba.

Within a few seconds the scheduler window will print:

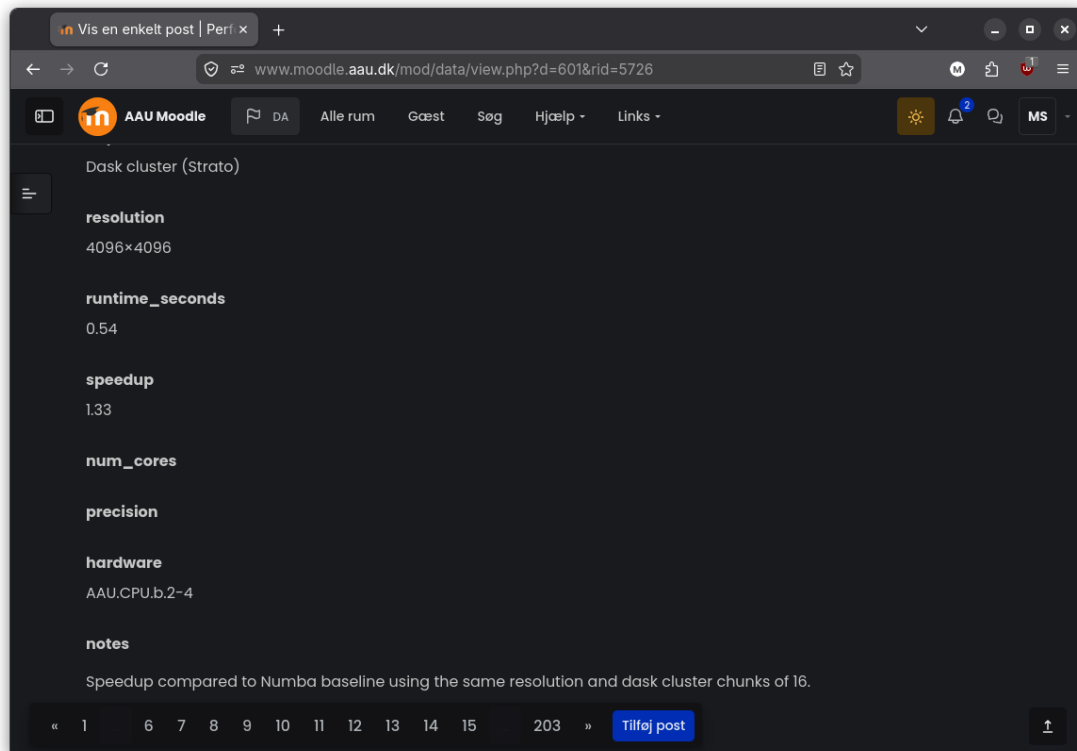
Register worker <Worker ...>

confirming the worker has connected. The dashboard Workers tab also updates immediately.

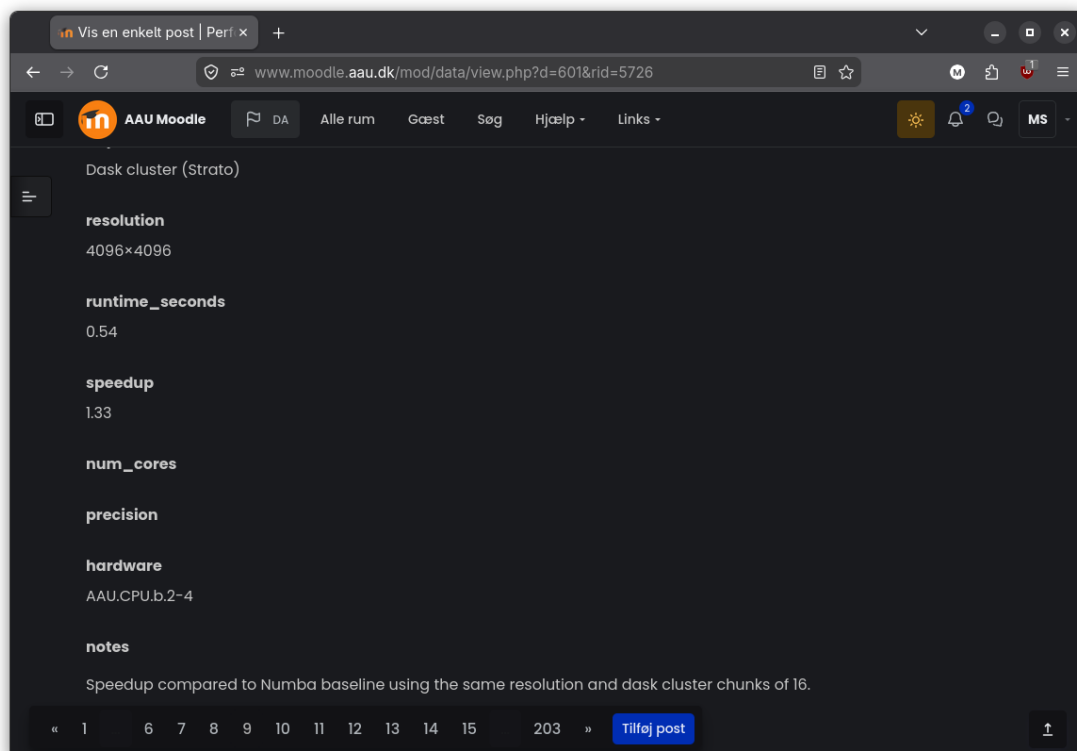
for 1 worker joined.



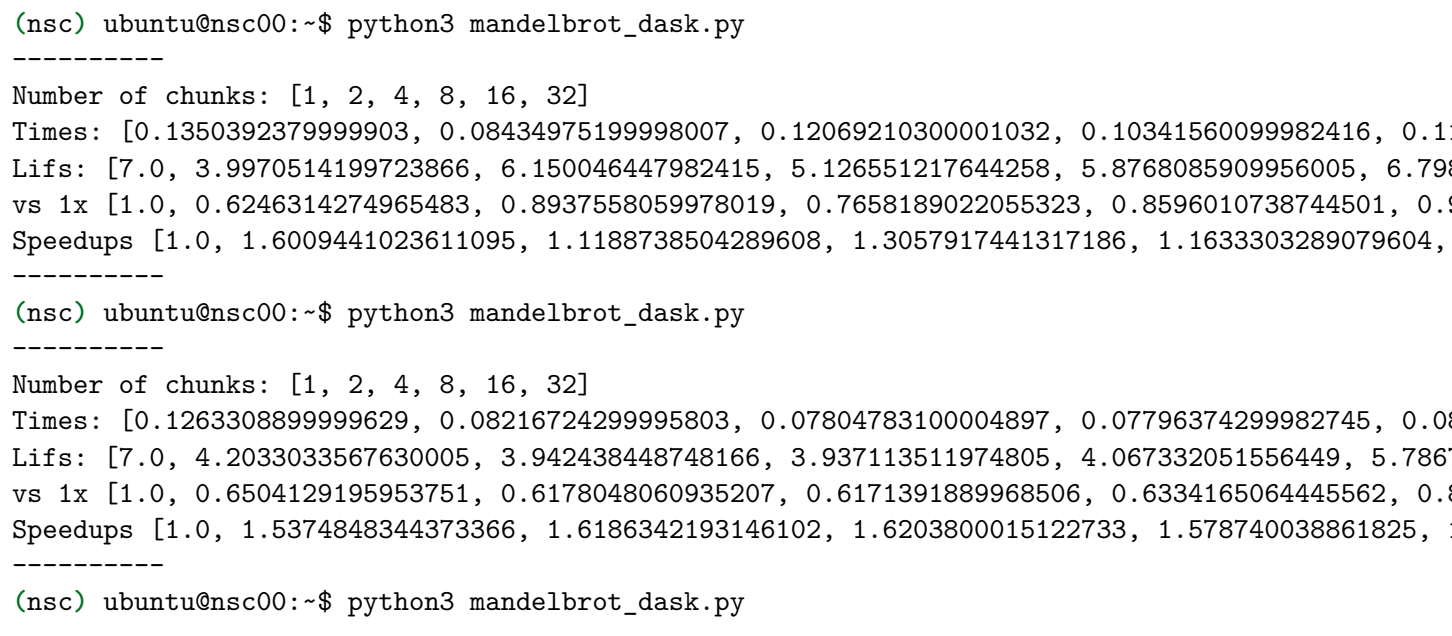
2 workers joined.



As the last worker number 3.



This is the output from one worker. I run my script on the main node noting what is prints, saving that to a csv file by hand.



\_\_\_\_\_

I have converted it to a csv file, so that it is easier to work with.

| workers | chunks | time   | lif  | vs1x  | speedup |
|---------|--------|--------|------|-------|---------|
| 1       | 1      | 0.135s | 7    | 1.00x | 1.00x   |
| 1       | 2      | 0.084s | 4    | 0.62x | 1.60x   |
| 1       | 4      | 0.121s | 6.15 | 0.89x | 1.12x   |
| 1       | 8      | 0.103s | 5.13 | 0.77x | 1.31x   |
| 1       | 16     | 0.116s | 5.88 | 0.86x | 1.16x   |
| 1       | 32     | 0.132s | 6.8  | 0.97x | 1.03x   |
| 2       | 1      | 0.126s | 7    | 1.00x | 1.00x   |
| 2       | 2      | 0.082s | 4.2  | 0.65x | 1.54x   |
| 2       | 4      | 0.078s | 3.94 | 0.62x | 1.62x   |
| 2       | 8      | 0.078s | 3.94 | 0.62x | 1.62x   |
| 2       | 16     | 0.080s | 4.07 | 0.63x | 1.58x   |
| 2       | 32     | 0.107s | 5.79 | 0.85x | 1.18x   |
| 3       | 1      | 0.129s | 7    | 1.00x | 1.00x   |
| 3       | 2      | 0.085s | 4.28 | 0.66x | 1.51x   |
| 3       | 4      | 0.077s | 3.76 | 0.60x | 1.68x   |
| 3       | 8      | 0.070s | 3.36 | 0.54x | 1.84x   |
| 3       | 16     | 0.083s | 4.14 | 0.64x | 1.56x   |
| 3       | 32     | 0.093s | 4.8  | 0.73x | 1.38x   |

If you prefer to work in a Jupyter notebook instead of running a script directly, see the [Appendix: Jupyter Notebook at the end of this guide](#).

Plain ssh is fine by me.

This is the output I get

```
(mikkel@pella)-[~/Sync/AAU/Semester 8/github/nsc-mikkel]
$ ssh -i nsc ubuntu@10.92.1.203
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Mar 30 20:32:35 UTC 2026

System load:  0.01               Processes:            104
Usage of /:   11.7% of 48.27GB    Users logged in:     1
Memory usage: 8%                 IPv4 address for ens3: 10.92.1.203
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

```
Last login: Mon Mar 30 19:50:39 2026 from 172.30.54.43
(base) ubuntu@nsc00:~$
```

#### 1.4.2 4.2 Run your script

```
python mandelbrot_dask.py
```

The output should include a line like:

```
<Client: 'tcp://10.92.y.x:8786' processes=6 threads=6, memory=11.46 GiB>
(base) ubuntu@nsc00:~$ conda activate nsc
(nsc) ubuntu@nsc00:~$ python mandelbrot_dask.py
```

I do not get the client print since I do not print it in my code.

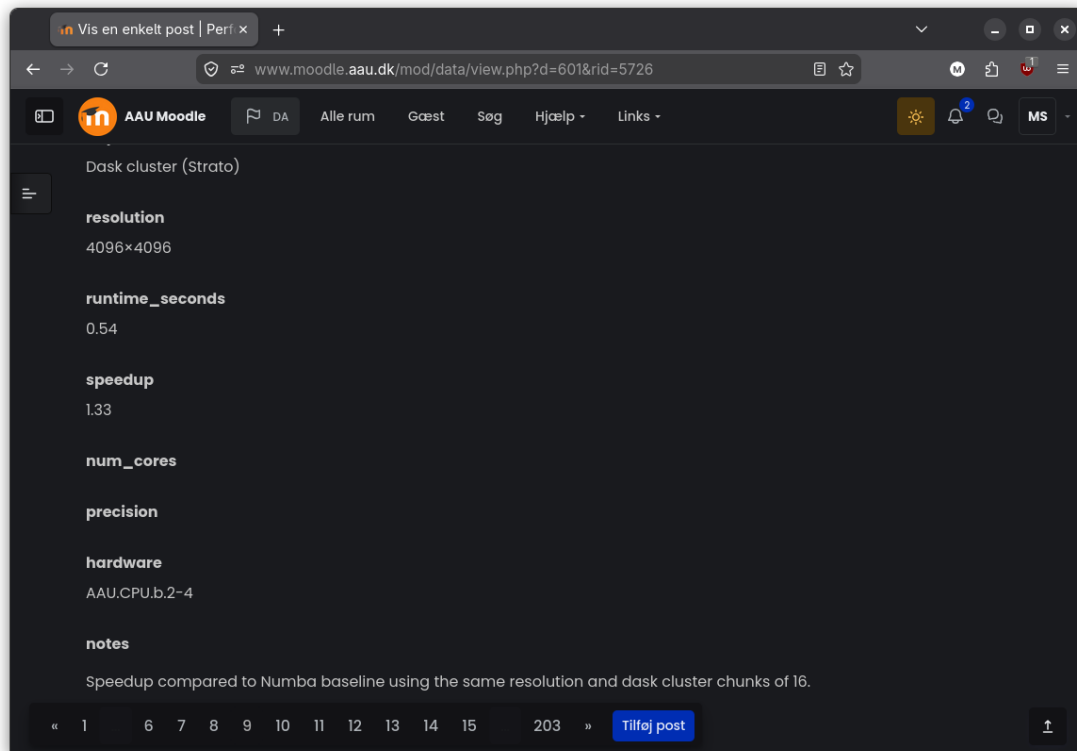
### 1.4.3 4.3 Check the dashboard

From your laptop browser (VPN required if off-campus):

<http://10.92.y.x:8787>

**You should see all workers listed under the Workers\*\* tab and tasks executing in real time.\*\***

I do see them!



### 1.4.4 4.4 Smoke test

Before benchmarking, verify all workers have the same package versions:

```
versions = client.run(lambda: __import__('dask').__version__)
print(versions)    # all values must be identical
```

If any worker shows a different version, delete the mismatched VM, re-launch from the snapshot, and reconnect.

In each VM in run

```
from dask.distributed import Client
client = Client()
versions = client.run(lambda: __import__('dask').__version__)
print(versions)
```

```
(nsc) ubuntu@nsc00:~$ python
Python 3.11.15 | packaged by conda-forge | (main, Mar 5 2026, 16:45:40) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from dask.distributed import Client
>>> client = Client()
/home/ubuntu/miniconda3/envs/nsc/lib/python3.11/site-packages/distributed/node.py:188: UserWarning:
Perhaps you already have a cluster running?
Hosting the HTTP server on port 40925 instead
  warnings.warn(
>>> versions = client.run(lambda: __import__('dask').__version__)
>>> print(versions)
{'tcp://127.0.0.1:35949': '2026.3.0', 'tcp://127.0.0.1:42887': '2026.3.0'}

(nsc) ubuntu@nsc-worker-1:~$ python
Python 3.11.15 | packaged by conda-forge | (main, Mar 5 2026, 16:45:40) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from dask.distributed import Client
>>> client = Client()
>>> versions = client.run(lambda: __import__('dask').__version__)
>>> print(versions)
{'tcp://127.0.0.1:33187': '2026.3.0', 'tcp://127.0.0.1:39931': '2026.3.0'}

(nsc) ubuntu@nsc-worker-2:~$ python
Python 3.11.15 | packaged by conda-forge | (main, Mar 5 2026, 16:45:40) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from dask.distributed import Client
>>> client = Client()
>>> versions = client.run(lambda: __import__('dask').__version__)
>>> print(versions)
{'tcp://127.0.0.1:38511': '2026.3.0', 'tcp://127.0.0.1:41779': '2026.3.0'}

(nsc) ubuntu@nsc-worker-3:~$ python
Python 3.11.15 | packaged by conda-forge | (main, Mar 5 2026, 16:45:40) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from dask.distributed import Client
>>> client = Client()
>>> versions = client.run(lambda: __import__('dask').__version__)
>>> print(versions)
{'tcp://127.0.0.1:38671': '2026.3.0', 'tcp://127.0.0.1:42203': '2026.3.0'}
```

They *all* have the same dask version.

## 1.5 Part 5 — Benchmark

Use  $N=4096^{**}$  or  $N=8192$  for meaningful speedup — at  $1024 \times 1024$  scheduling overhead dominates and speedup will be low.\*\*

\*\*\*\*Head node RAM:\*\* `dask.compute()` assembles the full result on the head node. A  $4096 \times 4096$  float64 grid is ~128 MB — fine on a small VM. At  $8192 \times 8192$  it is ~512 MB, and at  $16384 \times 16384$  it is ~2 GB. If you want to go beyond 8192, resize the head node to a larger flavor (e.g. AAU.CPU.b.4-8)

before running, otherwise the process may be killed by the OS. Workers do not need resizing — only the head node collects the result.\*\*

**Record your results in the performance tracker — include a Numba single-core run at the same resolution so speedup can be computed.**

I used  $N = 4096$ . I did the same as in part 3, saving the results to a csv file. Regarding the Numba case, I copied the code from lecture 4.

```
[5]: # From lecture 4

# From slide 45 (The updated once)

# mandelbrot_parallel.py (Tasks 1-3 are one continuous script)
import numpy as np
from numba import njit

@njit
def mandelbrot_pixel(c_real, c_imag, max_iter):
    z_real = z_imag = 0.0
    for i in range(max_iter):
        zr2 = z_real*z_real
        zi2 = z_imag*z_imag
        if zr2 + zi2 > 4.0: return i
        z_imag = 2.0*z_real*z_imag + c_imag
        z_real = zr2 - zi2 + c_real
    return max_iter

@njit
def mandelbrot_chunk(row_start, row_end, N,
                     x_min, x_max, y_min, y_max, max_iter):
    out = np.empty((row_end - row_start, N), dtype=np.int32)
    dx = (x_max - x_min) / N
    dy = (y_max - y_min) / N
    for r in range(row_end - row_start):
        c_imag = y_min + (r + row_start) * dy
        for col in range(N):
            out[r, col] = mandelbrot_pixel(x_min + col*dx, c_imag, max_iter)
    return out

def mandelbrot_serial(N, x_min, x_max, y_min, y_max, max_iter=100):
    return mandelbrot_chunk(0, N, N, x_min, x_max, y_min, y_max, max_iter)
```

```
[6]: import statistics
import time

times = []
for _ in range(3):
```

```

t0 = time.perf_counter()
mandelbrot_serial(4096, -2.5, 1.0, -1.25, 1.25)
times.append(time.perf_counter() - t0)
print(statistics.median(times))

```

0.71728592099862

```

[9]: from IPython.display import Markdown, display
import pandas as pd

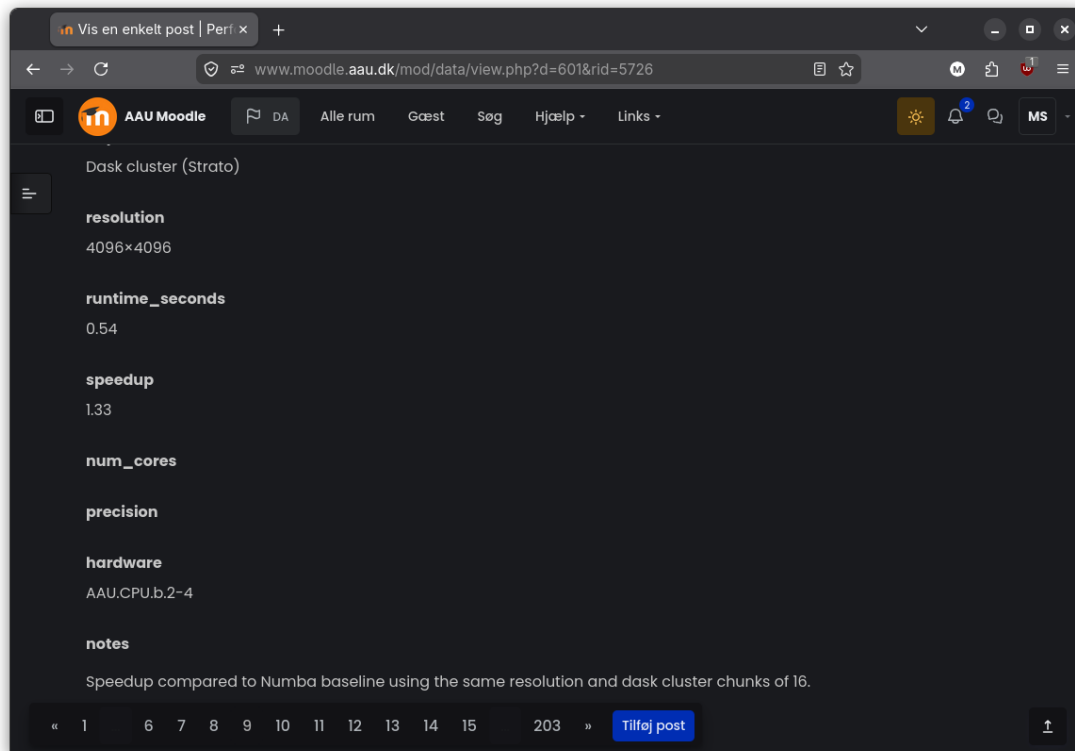
df = pd.read_csv("l07_part5.csv")

# Compute speedups
t0 = 0.71728592099862
df["speedup"] = t0 / df["time"]

# Display as table
df["time"] = df["time"].map("{:>.3f}s".format)
df["speedup"] = df["speedup"].map("{:>.2f}x".format)
display(Markdown(df[["workers", "chunks", "time", "speedup"]].
    ↪to_markdown(index=False)))

```

| workers | chunks | time   | speedup |
|---------|--------|--------|---------|
| 1       | 1      | 1.646s | 0.44x   |
| 1       | 2      | 0.907s | 0.79x   |
| 1       | 4      | 0.918s | 0.78x   |
| 1       | 8      | 1.011s | 0.71x   |
| 1       | 16     | 1.011s | 0.71x   |
| 1       | 32     | 0.984s | 0.73x   |
| 2       | 1      | 1.527s | 0.47x   |
| 2       | 2      | 0.933s | 0.77x   |
| 2       | 4      | 0.858s | 0.84x   |
| 2       | 8      | 0.758s | 0.95x   |
| 2       | 16     | 0.646s | 1.11x   |
| 2       | 32     | 0.598s | 1.20x   |
| 3       | 1      | 1.700s | 0.42x   |
| 3       | 2      | 0.951s | 0.75x   |
| 3       | 4      | 0.841s | 0.85x   |
| 3       | 8      | 0.873s | 0.82x   |
| 3       | 16     | 0.540s | 1.33x   |
| 3       | 32     | 0.559s | 1.28x   |



## 1.6 Part 6 — Clean Up When Done

Running VMs consume your quota even when idle.

1. **\*\*\*\*Worker VMs:\*\* Delete them** — you can always re-launch from the snapshot in minutes. In **Compute** → **Instances**, select each worker, click **Delete Instances**.**\*\***
2. **\*\*\*\*Head node:\*\* You have two options:\*\***
  - **\*\*\*\*Shelve (recommended):\*\*** frees the vCPU/RAM quota but keeps the disk. Resume later with Unshelve — faster than re-launching. Click the dropdown → **Shelve Instance**.**\*\***
  - **\*\*\*\*Delete:\*\*** removes everything. Only do this if you are certain you no longer need it. Your snapshot still exists and you can re-launch from it.**\*\***
3. **\*\*\*\*Snapshot:\*\* Keep it.** Snapshots use storage quota (not compute quota) and let you rebuild the cluster in minutes next time.**\*\***

I have chosen to just delete the instances but keep their virtual disks.

## 1.7 Troubleshooting

**\*\*\*\*Worker won't connect to scheduler\*\*\*\*** - Check that port 8786 is open in the security group (Part 1.1). - Confirm you are using the head node's IP from the Instances list (e.g. 10.92.y.x). - Check the scheduler is still running: `tmux attach -t dask` on the head node.

**\*\*\*\*Dashboard not loading in browser\*\*\*\*** - Confirm port 8787 is open in the security group. - Confirm VPN is connected. - Try `http://10.92.y.x:8787` (not https).

\*\*\*\*SSH connection drops / scheduler stops\*\*\*\* - Always use `tmux` for the scheduler (Part 3.2). If you forgot and it stopped, just re-run `dask scheduler` and reconnect the workers.

\*\*\*\*VPN not working at AAU\*\*\*\* - Use your laptop as a mobile hotspot (share from your phone). - Connect both your laptop and any physical co-worker laptops to that hotspot. - Use internal IPs as normal — devices on the same hotspot subnet can reach each other directly. - This also works if you want to form a temporary cluster using classmates' laptops as additional workers: each person runs `dask worker YOUR_IP:8786` on their own machine.

\*\*\*\*Version mismatch between workers\*\*\*\* - Always launch workers from the snapshot, not from the base Ubuntu image. - Run the smoke test (Part 4.5) to catch mismatches before benchmarking.

## 1.8 Appendix: Jupyter Notebook

If you prefer working in a notebook rather than running a script, you can run Jupyter on the head node and tunnel it to your laptop browser.

\*\*\*\*On the head node:\*\*\*\*

```
conda activate nsc
jupyter notebook --no-browser --port=8888
```

\*\*\*\*On your laptop (in a separate terminal):\*\*\*\*

```
ssh -i ~/.ssh/my_ssh_key -L 8888:localhost:8888 ubuntu@10.92.y.x -N
```

Then open `http://localhost:8888` in your browser.

The three code changes in Part 4.2 apply equally to a notebook — replace the `LocalCluster` cell with `client = Client("tcp://10.92.y.x:8786")` and remove any `cluster.close()` call.